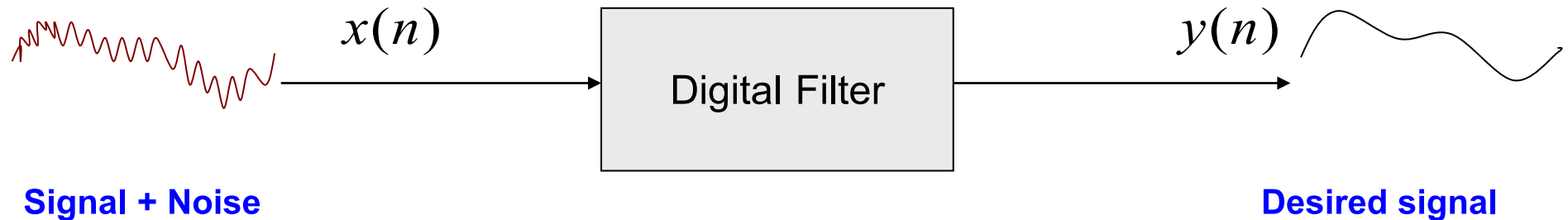


7

Digital Filter Structures

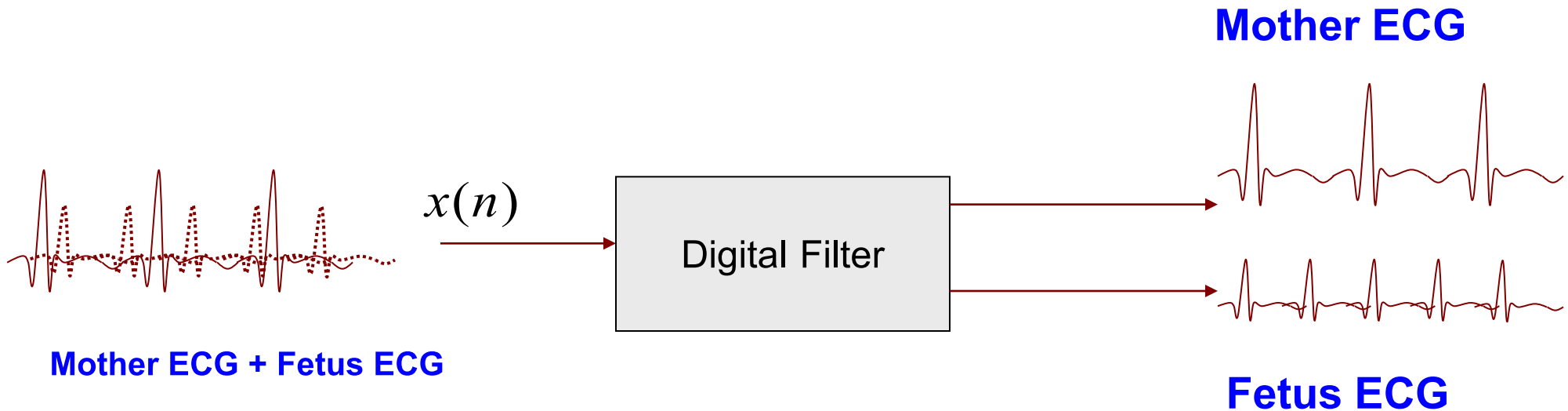
Applications of Digital Filters

- Digital filters are processing of time domain signals resulting in the desired frequency response.



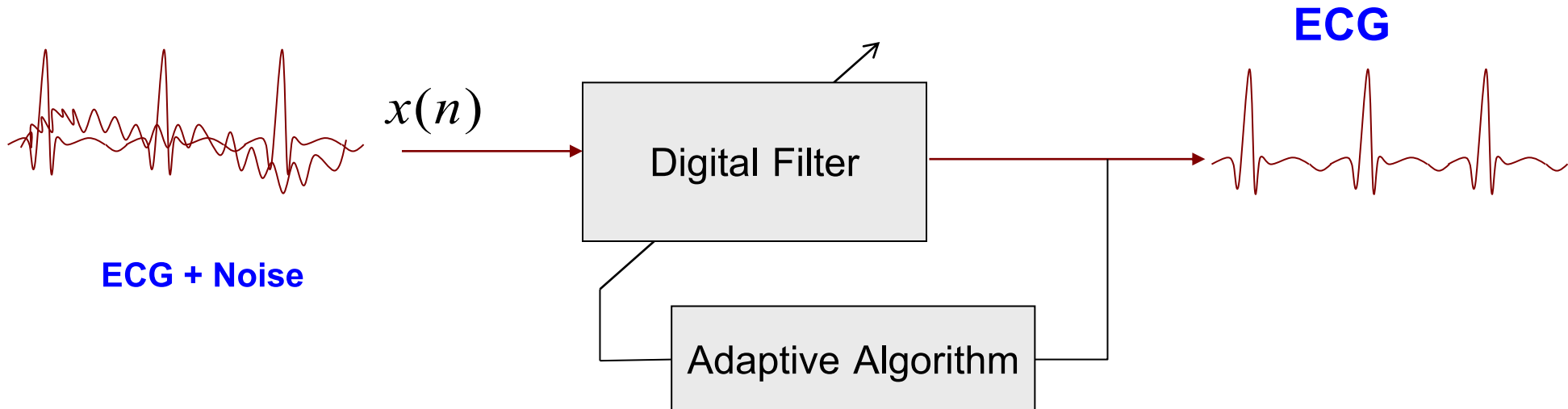
Frequency response overlapping with noise.

- Digital filters can be used even when the desired signal has frequency response overlapping with noise.

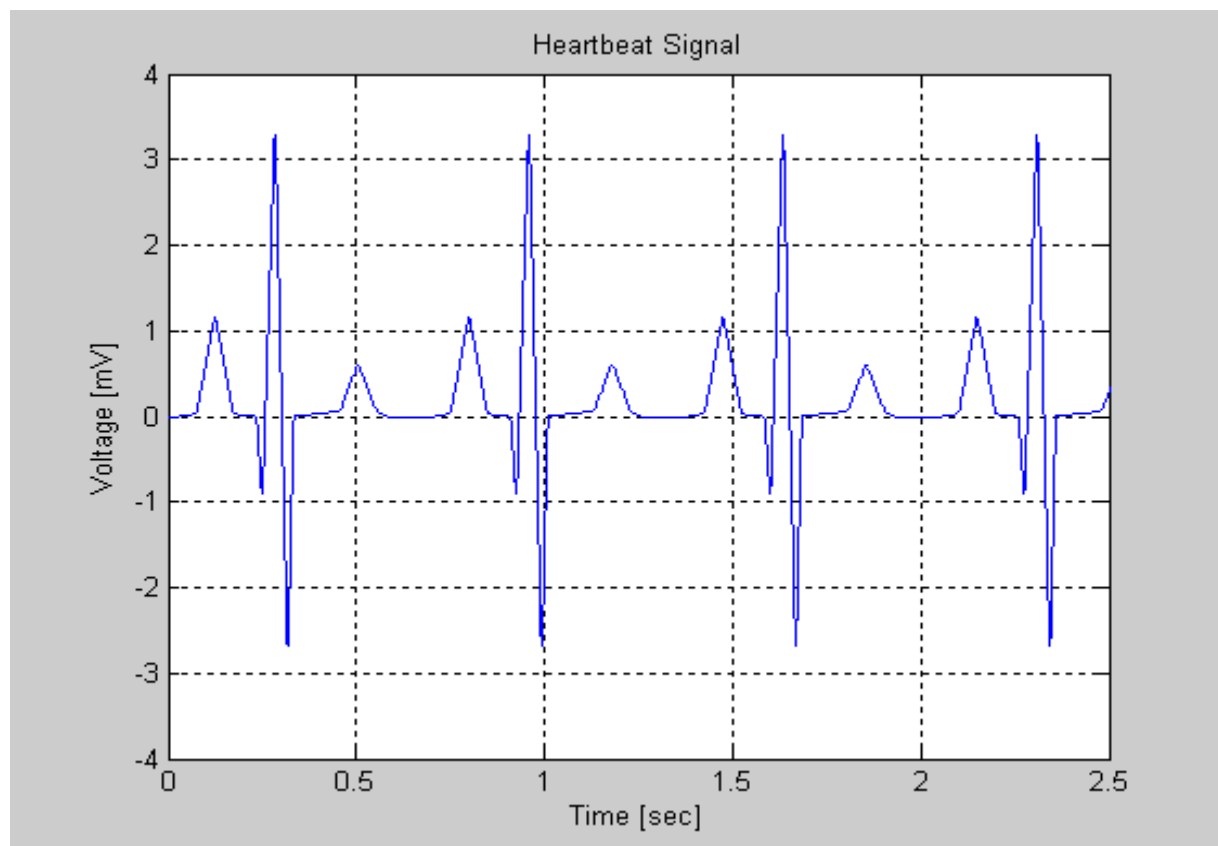


Adaptive Filters

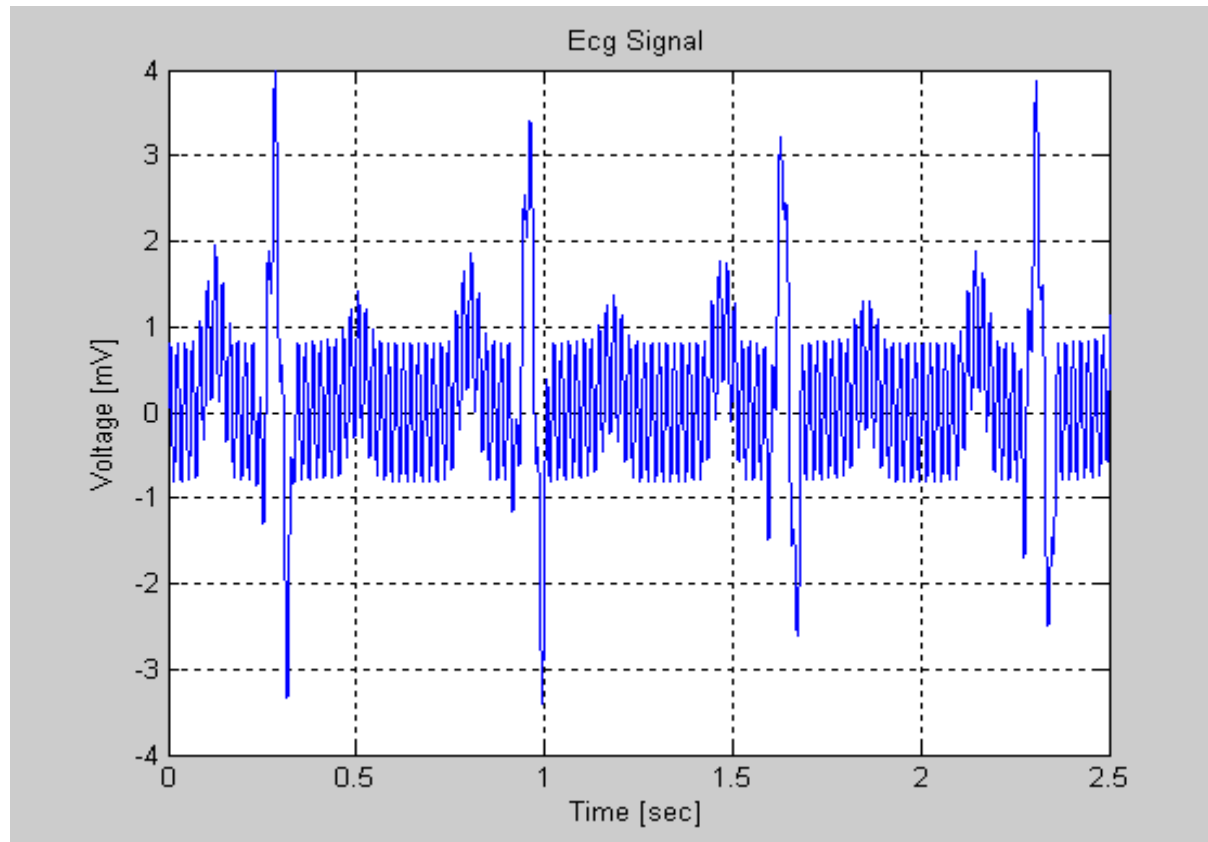
- When there is frequency response overlapping, an adaptive filter can be used. However, the theory of operation is different from the basic digital filters.



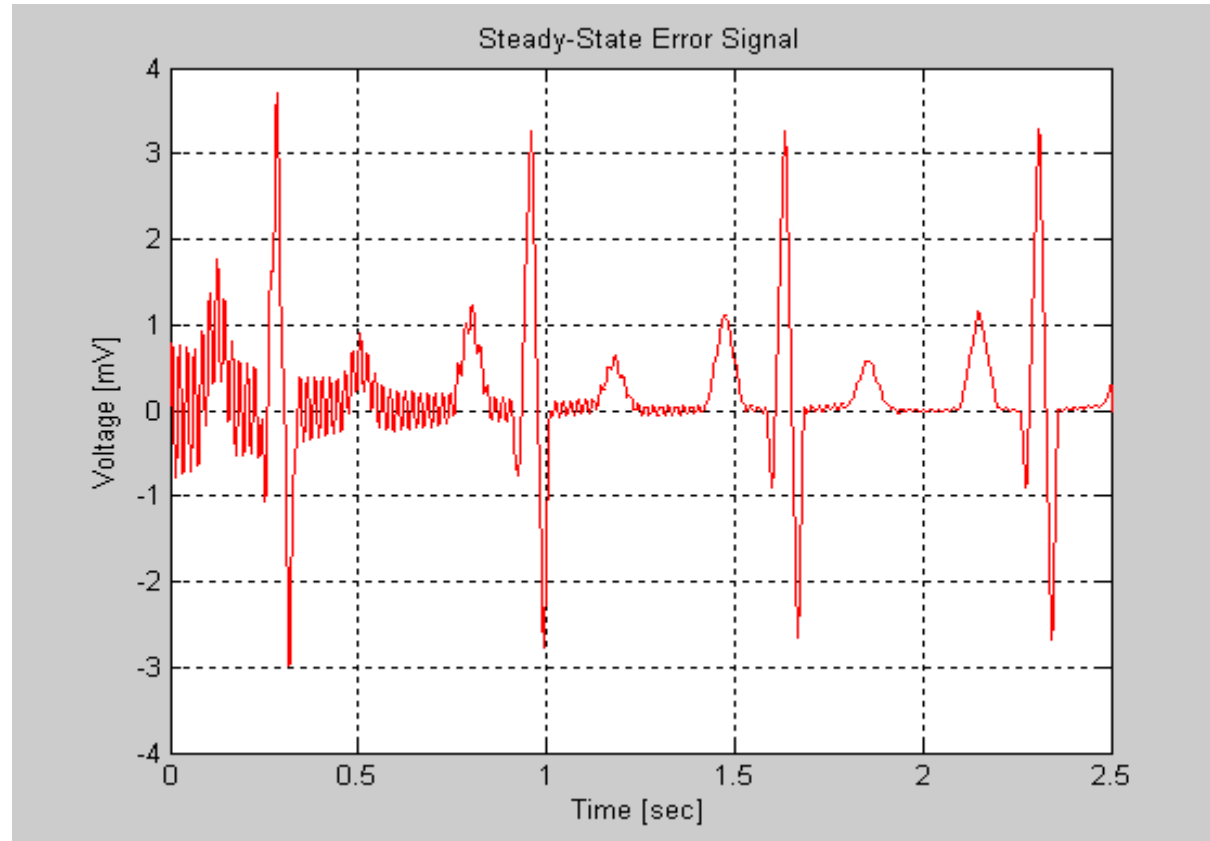
ECG



ECG with 50 Hz Noise



ECG with **50 Hz** Noise Being Eliminated using Adaptive Filtering



Types of Digital Filters

- From the difference equation

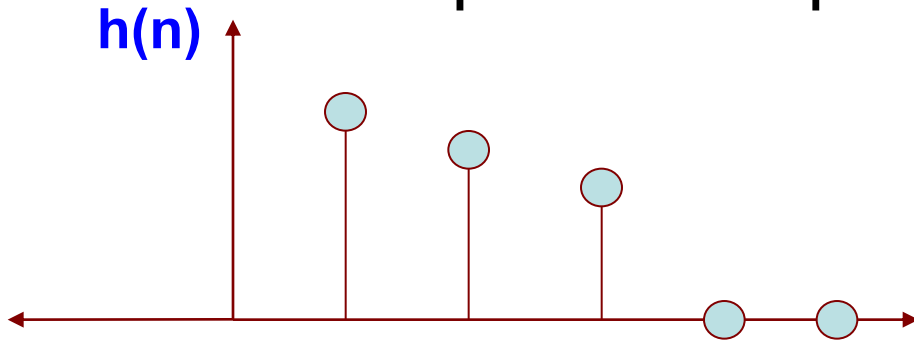
$$y(n) = -a(1)y(n-1) + b(0)x(n) + b(1)x(n-1)$$

we can construct the transfer function in the form of

$$H(z) = \frac{B(z)}{A(z)} = \frac{\sum_{k=0}^q b(k)z^{-k}}{1 + \sum_{k=1}^p a(k)z^{-k}}$$

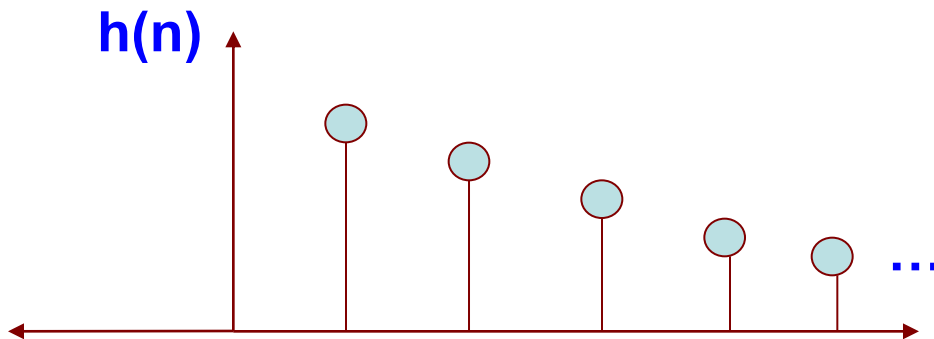
FIR and IIR

- Finite Impulse Response (FIR)



$$H(z) = B(z) = \sum_{k=0}^q b(k)z^{-k}$$

- Infinite Impulse Response: IIR

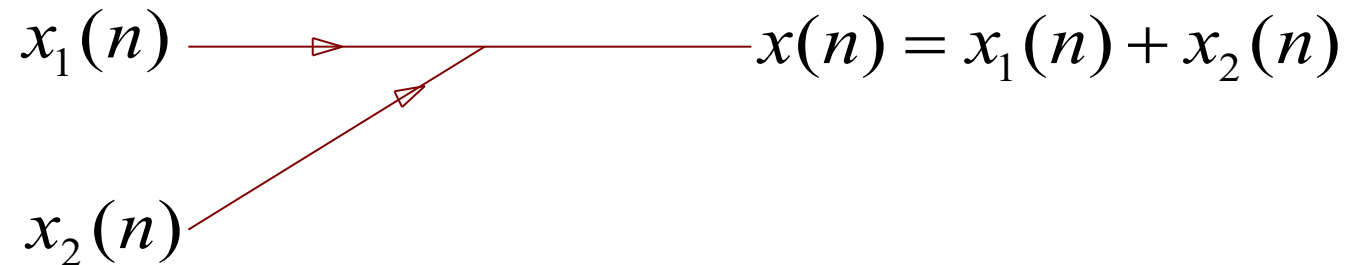


$$H(z) = \frac{B(z)}{A(z)} = \frac{\sum_{k=0}^q b(k)z^{-k}}{1 + \sum_{k=1}^p a(k)z^{-k}}$$

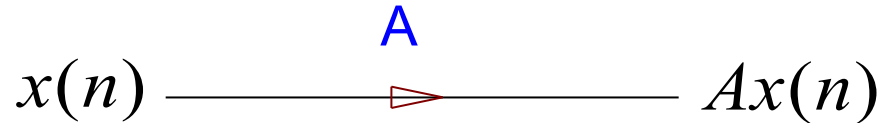
Basic Block Diagrams

- Basic Elements

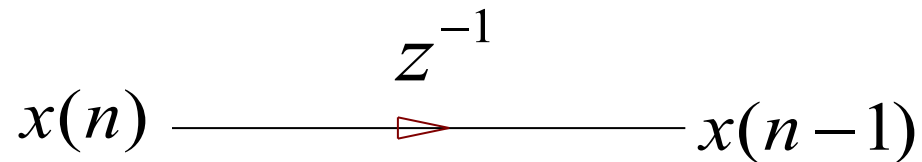
- Adder



- Multiplier



- Delay



Example 7.1

- Determine the z-Transform for the transfer function $H(z)$ of the following difference equations

$$a) \quad y(n) = 2x(n) + x(n-1)$$

$$b) \quad y(n) = 2x(n) + 0.5x(n-4)$$

$$c) \quad y(n) = 0.8y(n-1) + 2x(n) + x(n-1)$$

$$a) \quad y(n) = 2x(n) + x(n-1)$$

- Perform z-transform

$$\begin{aligned} Y(z) &= 2X(z) + z^{-1}X(z) \\ &= (2 + z^{-1})X(z) \end{aligned}$$

$$b) \quad y(n) = 2x(n) + 0.5x(n-2)$$

- Perform z-transform

$$\begin{aligned} Y(z) &= 2X(z) + 0.5z^{-2}X(z) \\ &= (2 + 0.5z^{-2})X(z) \end{aligned}$$

$$c) \quad y(n) = 0.8y(n-1) + 2x(n) + x(n-1)$$

$$\begin{aligned} Y(z) &= 0.8z^{-1}Y(z) + 2X(z) + z^{-1}X(z) \\ &= 0.8z^{-1}Y(z) + (2 + z^{-1})X(z) \end{aligned}$$

Example 7.2

Sketch the signal flow graph of the following difference equations

$$a) \quad y(n) = 2x(n) + x(n-1)$$

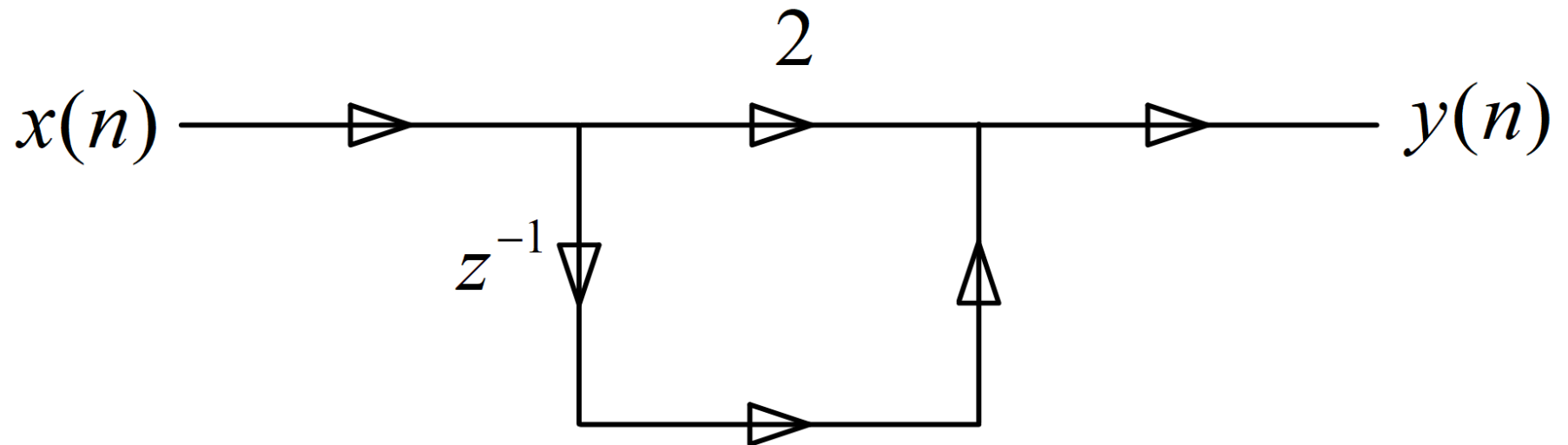
$$b) \quad y(n) = 2x(n) + 0.5x(n-2)$$

$$c) \quad y(n) = 0.8y(n-1) + 2x(n) + x(n-1)$$

$$a) \quad y(n] = 2x(n) + x(n - 1)$$

$$Y(z) = 2X(z) + z^{-1}X(z)$$

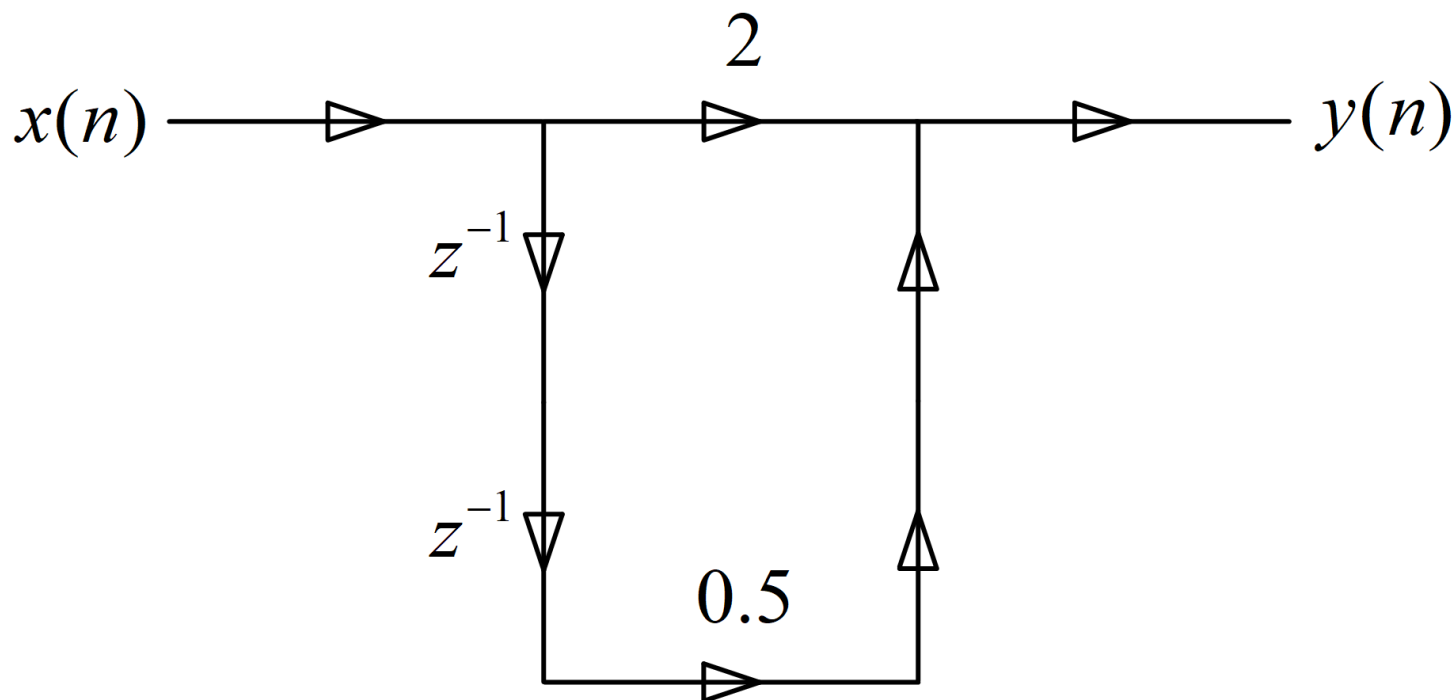
$$= (2 + z^{-1})X(z)$$



$$b) \quad y(n] = 2x(n) + 0.5x(n - 2)$$

$$Y(z) = 2X(z) + 0.5z^{-2}X(z)$$

$$= (2 + 0.5z^{-2})X(z)$$



$$c) \quad y(n) = 0.8y(n-1) + 2x(n) + x(n-1)$$

$$\begin{aligned} Y(z) &= 0.8z^{-1}Y(z) + 2X(z) + z^{-1}X(z) \\ &= 0.8z^{-1}Y(z) + (2 + z^{-1})X(z) \end{aligned}$$

$$Y(z) = 0.8z^{-1}Y(z) + 2X(z) + z^{-1}X(z)$$

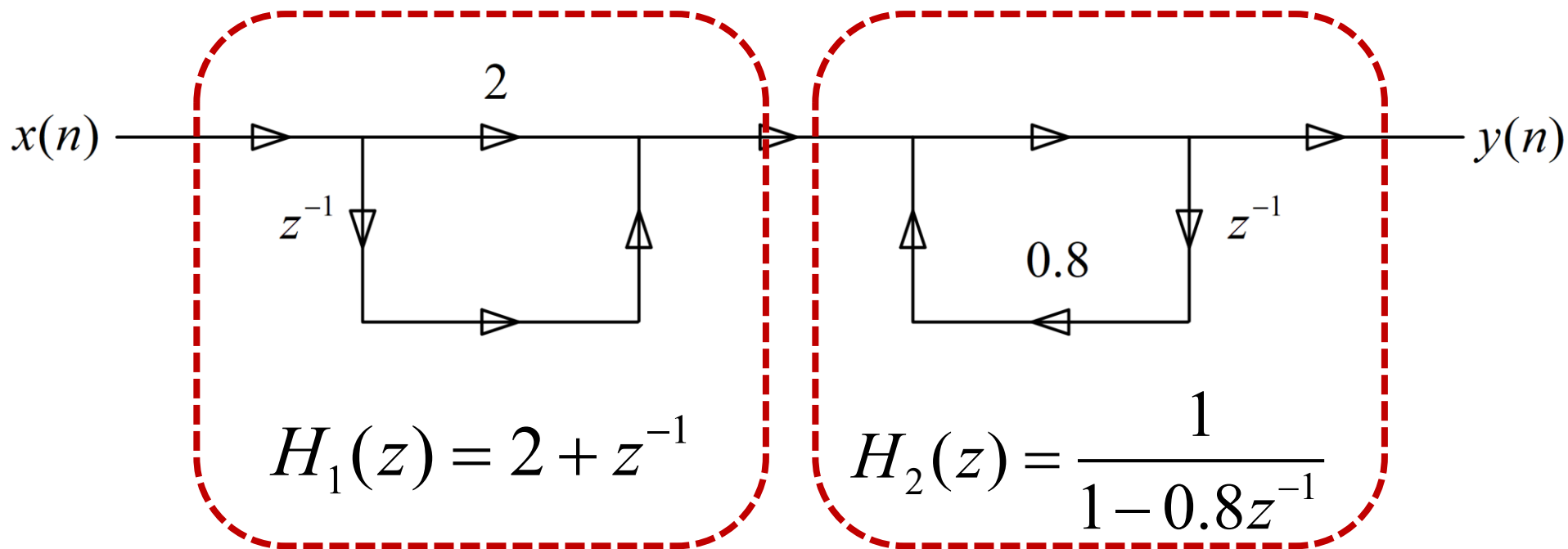
$$Y(z) = 0.8z^{-1}Y(z) + (2 + z^{-1})X(z)$$

$$(1 - 0.8z^{-1})Y(z) = (2 + z^{-1})X(z)$$

$$\frac{Y(z)}{X(z)} = \frac{2 + z^{-1}}{1 - 0.8z^{-1}}$$

$$H(z) = 2 + z^{-1} \cdot \frac{1}{1 - 0.8z^{-1}}$$

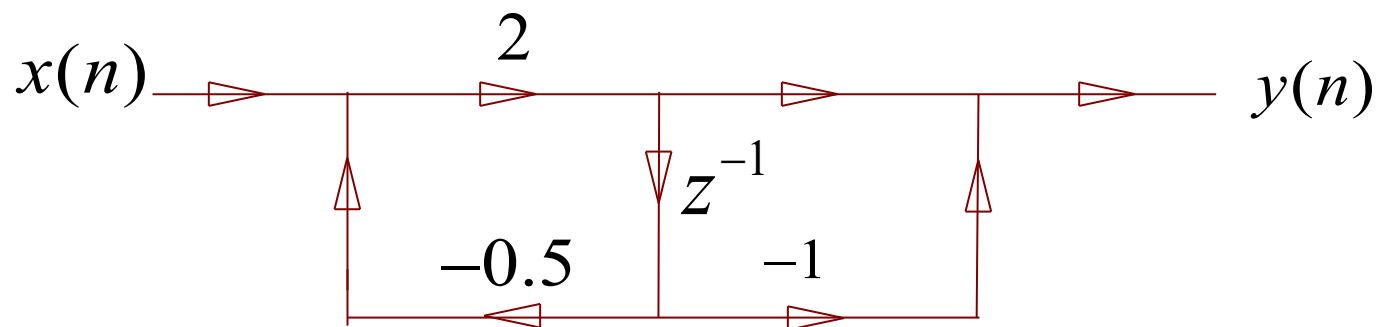
$$H(z) = H_1(z)H_2(z)$$

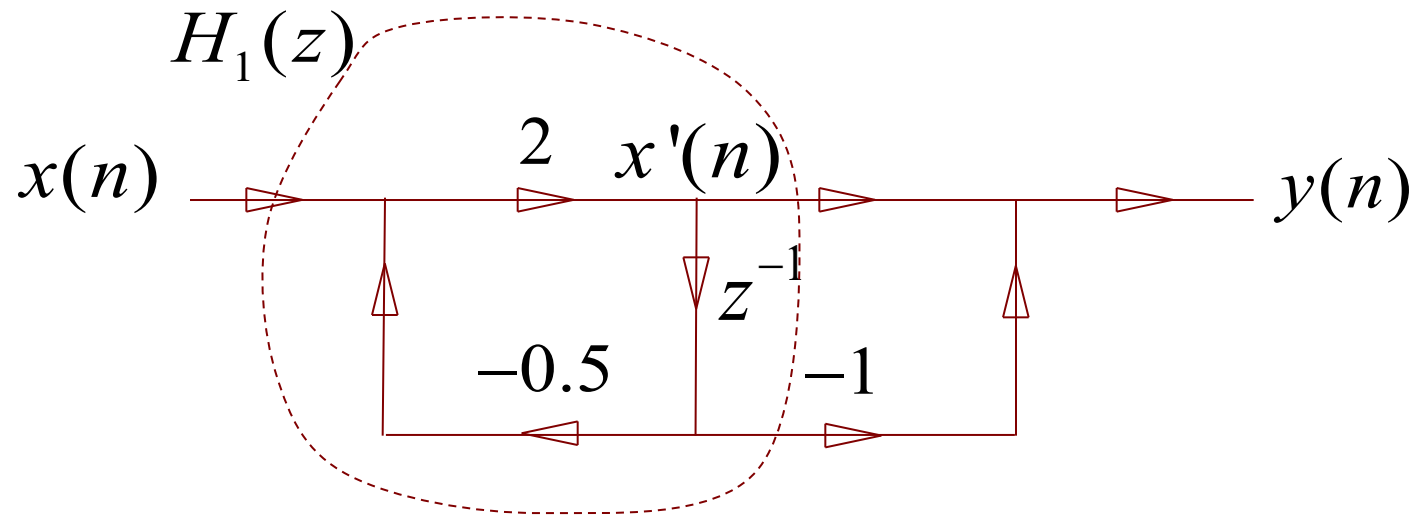


$$H(z) = H_1(z)H_2(z)$$

Example 7.3

- Find $H(z)$ from the direct form II structure below



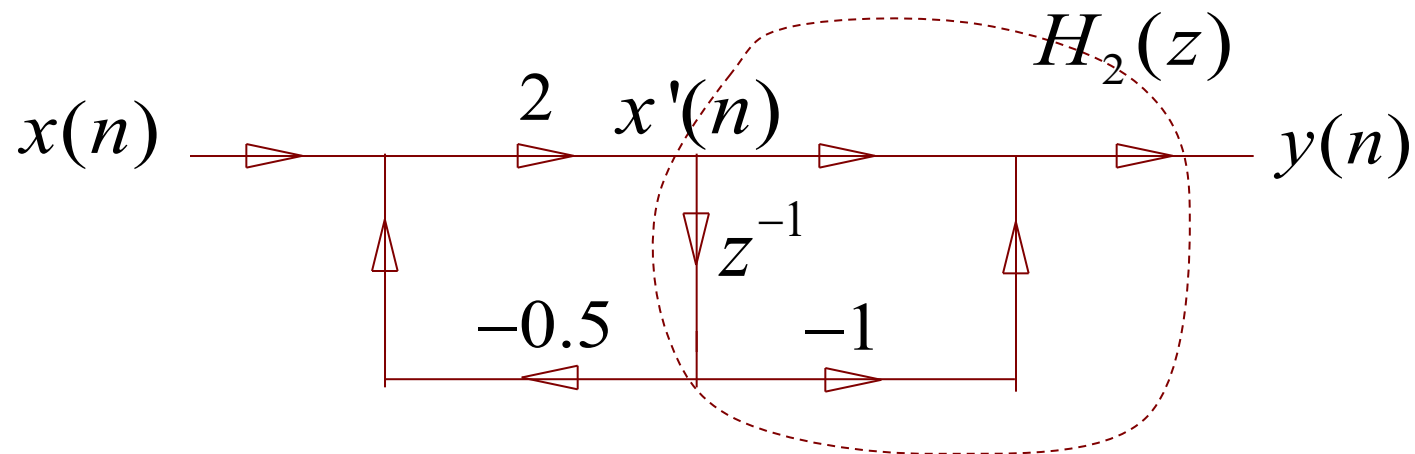


$$x'(n) = 2(x(n) - 0.5x'(n-1))$$

$$X'(z) = 2(X(z) - 0.5z^{-1}X'(z))$$

$$X'(z) + z^{-1}X'(z) = 2X(z)$$

$$H_1(z) = \frac{X'(z)}{X(z)} = \frac{2}{1 + z^{-1}}$$



$$y(n) = x'(n) - x'(n-1)$$

$$Y(z) = X'(z) - z^{-1}X'(z)$$

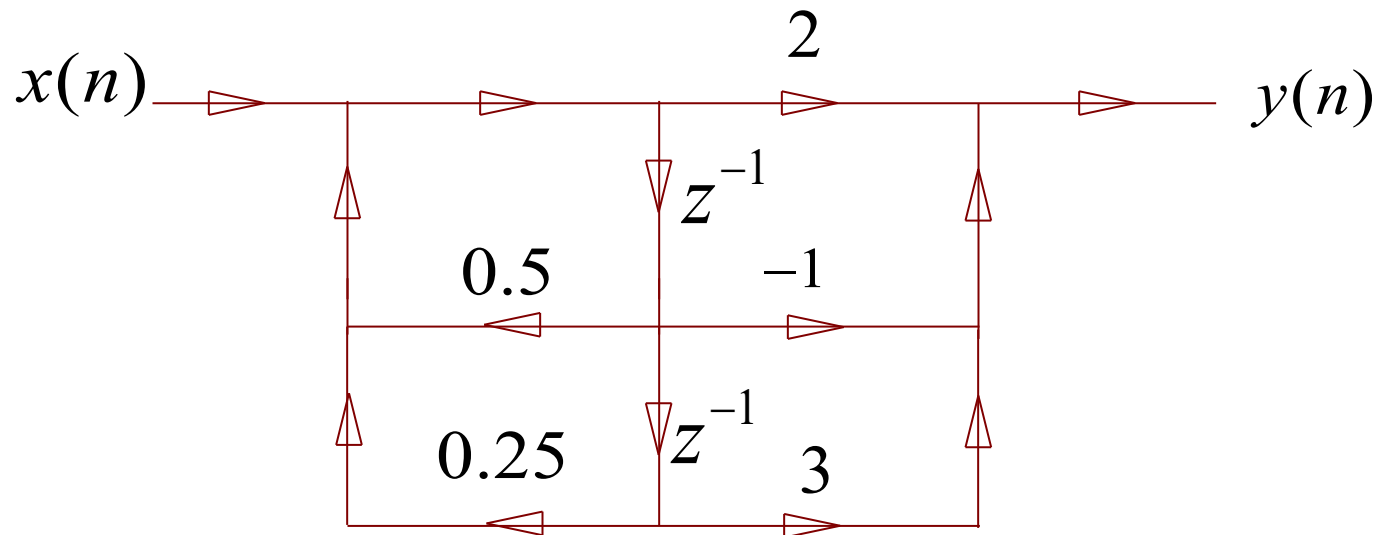
$$H_2(z) = \frac{Y(z)}{X'(z)} = 1 - z^{-1}$$

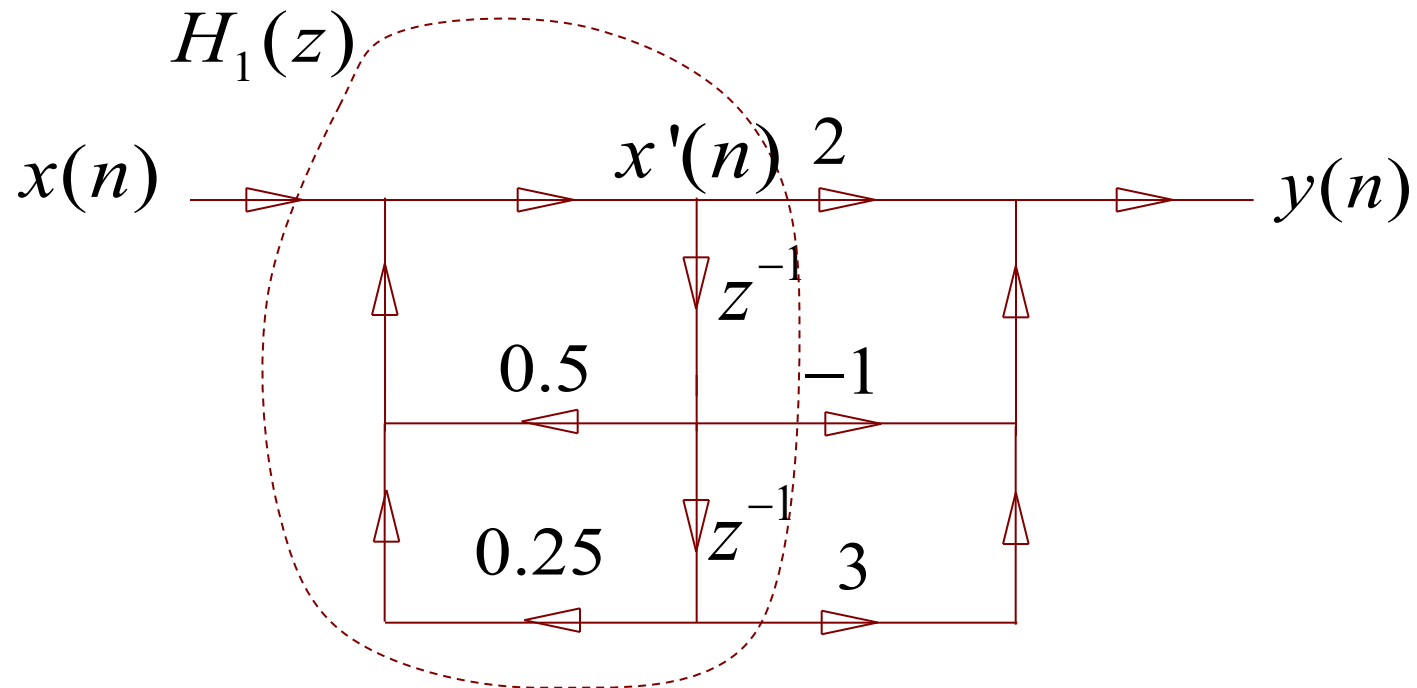
- We have the overall transfer function of

$$\begin{aligned} H(z) &= H_1(z)H_2(z) \\ &= \frac{X'(z)}{X(z)} \frac{Y(z)}{X'(z)} \\ &= \frac{2 - 2z^{-1}}{1 + z^{-1}} \end{aligned}$$

QUIZ 7.1

- Find $H(z)$ from the direct form II structure below



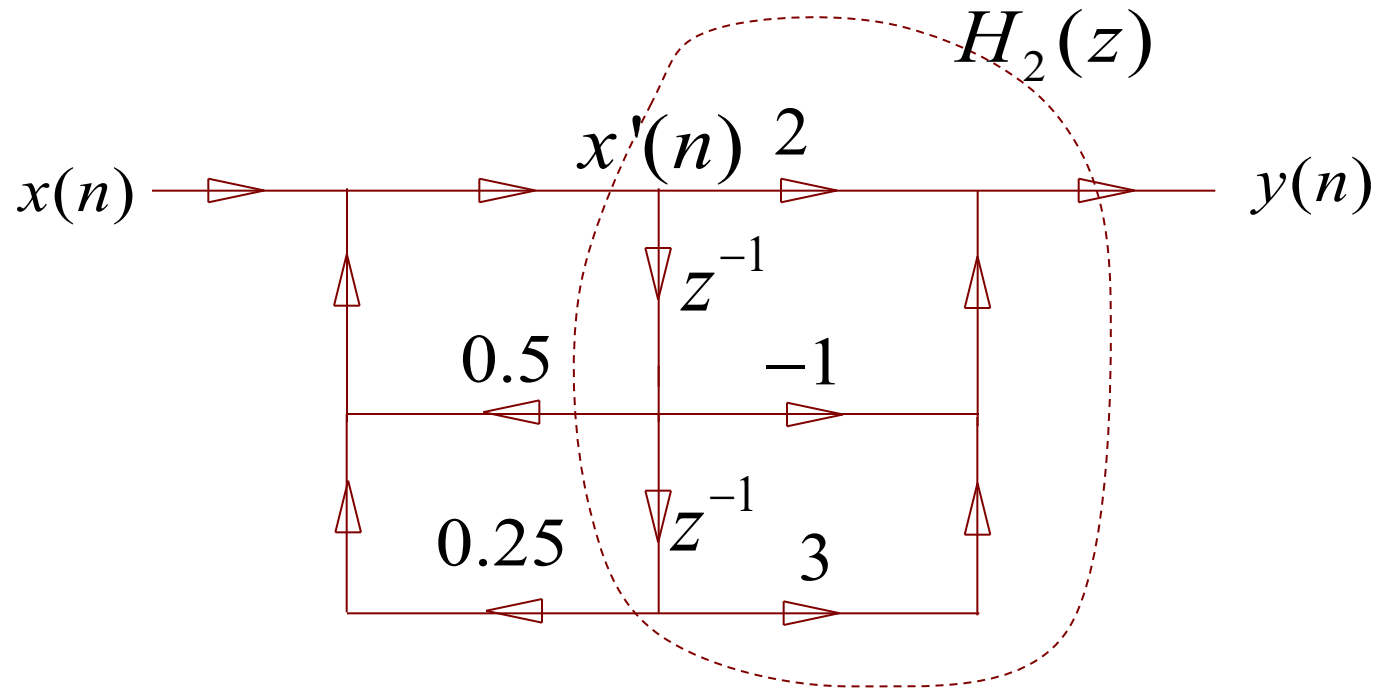


$$x'(n) = x(n) + 0.5x'(n-1) + 0.25x'(n-2)$$

$$X'(z) = X(z) + 0.5z^{-1}X'(z) + 0.25z^{-2}X'(z)$$

$$X'(z) - 0.5z^{-1}X'(z) - 0.25z^{-2}X'(z) = X(z)$$

$$H_1(z) = \frac{X'(z)}{X(z)} = \frac{1}{1 - 0.5z^{-1} - 0.25z^{-2}}$$



$$y(n) = 2x'(n) - x'(n-1) + 3x'(n-2)$$

$$Y(z) = 2X'(z) - z^{-1}X'(z) + 3z^{-2}X'(z)$$

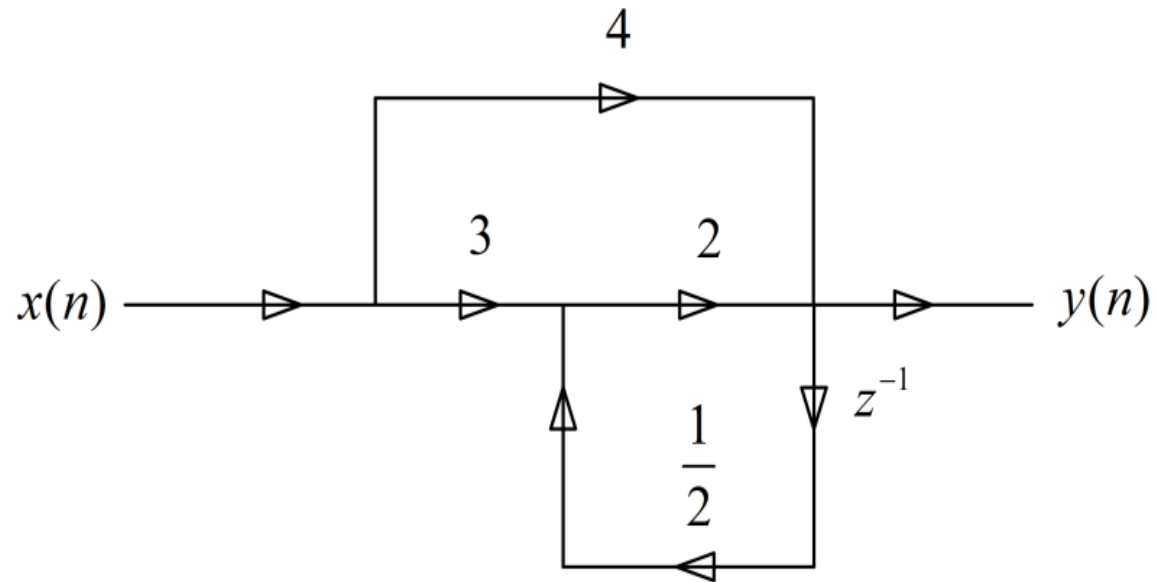
$$H_2(z) = \frac{Y(z)}{X'(z)} = 2 - z^{-1} + 3z^{-2}$$

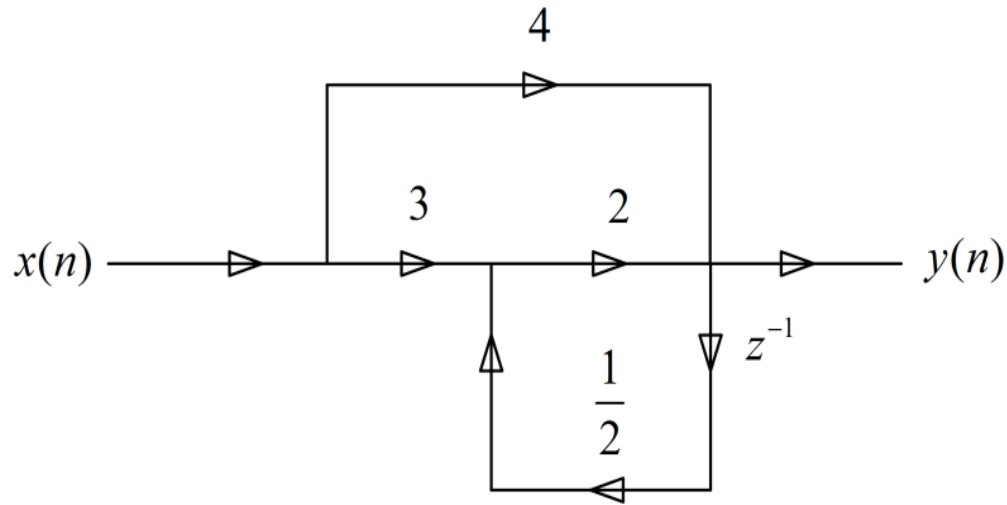
- We have the overall transfer function of

$$\begin{aligned} H(z) &= H_1(z)H_2(z) \\ &= \frac{2 - z^{-1} + 3z^{-2}}{1 - 0.5z^{-1} - 0.25z^{-2}} \end{aligned}$$

Example 7.4

- Determine $H(z)$ from the structure below





$$v(n) = 3x(n) + \frac{1}{2} y(n-1)$$

$$y(n) = 2v(n) + 4x(n)$$

$$y(n) = 2\left[3x(n) + \frac{1}{2} y(n-1)\right] + 4x(n)$$

$$= 6x(n) + y(n-1) + 4x(n)$$

$$= 10x(n) + y(n-1)$$

$$y(n) - y(n-1) = 10x(n)$$

$$Y(z) - z^{-1}Y(z) = 10X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{10}{1 - z^{-1}}$$

Using the z-transform pairs table to convert $H(z)$ to $h(n)$

Table 1 z-tranform pairs

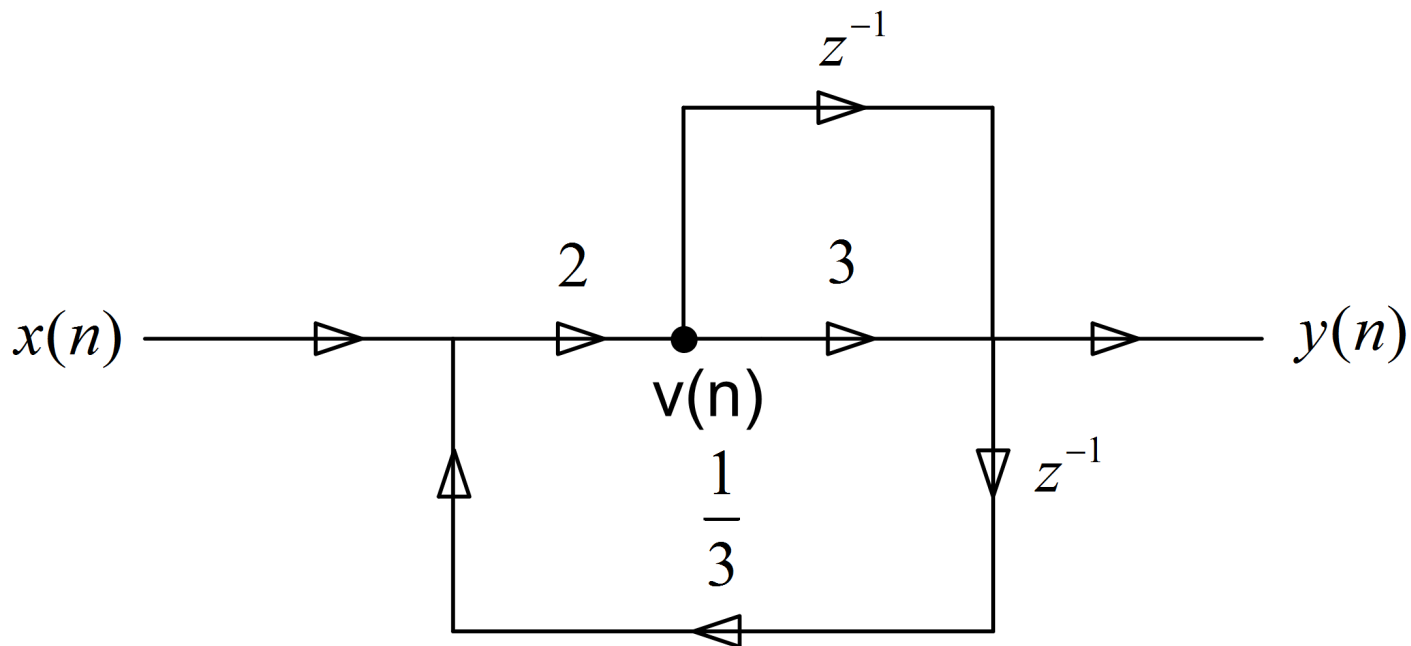
Sequence $x(n)$	z-transform $X(z)$	Region of Convergence (ROC)
$\delta(n)$	1	All z
$u(n)$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
$a^n u(n)$	$\frac{1}{1 - az^{-1}}$	$ z > a $
$-a^n u(-n-1)$	$\frac{1}{1 - az^{-1}}$	$ z < a $
$2 C p ^n \cos(\omega_0 n + \theta)u(n)$	$\frac{C}{1 - pz^{-1}} + \frac{C^*}{1 - p^* z^{-1}}$	$ z > p $

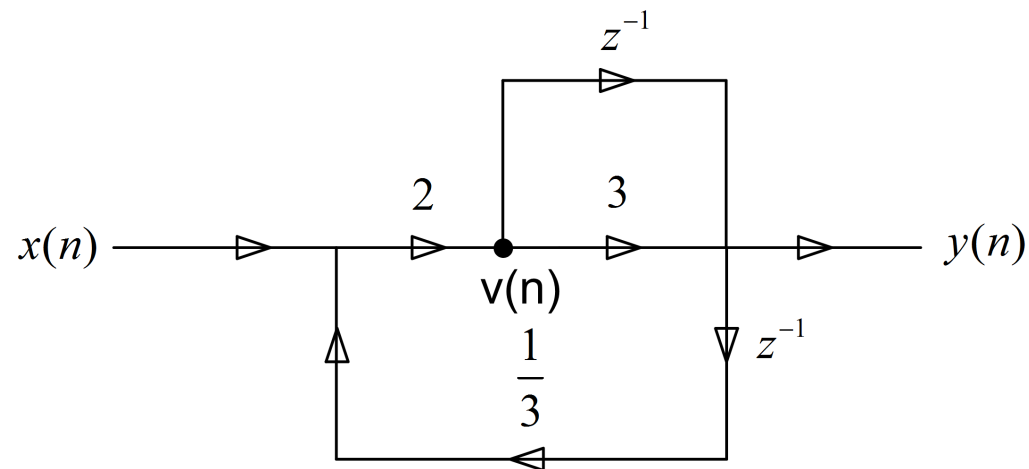
Impulse Response $h(n)$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{10}{1 - z^{-1}}$$

$$h(n) = 10u(n)$$

Example 7.6





$$v(n) = 2 \left(x(n) + \frac{1}{3} y(n-1) \right)$$

$$y(n) = v(n-1) + 3v(n)$$

$$V(z) = 2 \left(X(z) + \frac{1}{3} z^{-1} Y(z) \right)$$

$$\begin{aligned}
 Y(z) &= z^{-1}V(z) + 3V(z) \\
 &= (3 + z^{-1})V(z)
 \end{aligned}$$

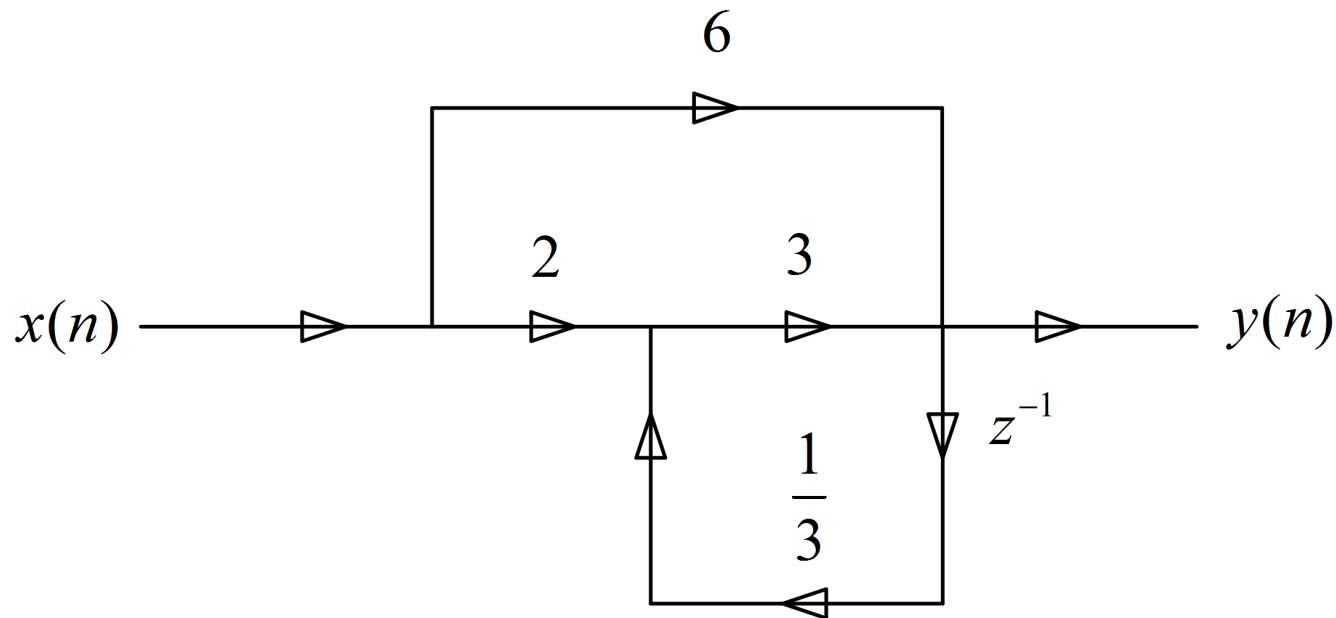
$$\begin{aligned}
 Y(z) &= (3 + z^{-1})V(z) \\
 &= (3 + z^{-1}) \cdot 2 \left(X(z) + \frac{1}{3} z^{-1} Y(z) \right) \\
 &= (6 + 2z^{-1})X(z) + \left(2z^{-1} + \frac{2}{3} z^{-2} \right) Y(z)
 \end{aligned}$$

$$\left(1 - 2z^{-1} - \frac{2}{3}z^{-2}\right)Y(z) = (6 + 2z^{-1})X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{6 + 2z^{-1}}{1 - 2z^{-1} - \frac{2}{3}z^{-2}}$$

Homework 7.1

- Determine transfer function $H(z)$ and Impulse Response $h(n)$ of the filter structure below



$$S_1 : \quad v(n) = 2x(n) + \frac{1}{3}y(n-1)$$

$$S_2 : \quad y(n) = 3v(n) + 6x(n)$$

$$\begin{aligned} y(n) &= 3\left[2x(n) + \frac{1}{3}y(n-1)\right] + 6x(n) \\ &= 6x(n) + y(n-1) + 6x(n) \\ &= 12x(n) + y(n-1) \end{aligned}$$

$$y(n) - y(n-1) = 12x(n)$$

$$Y(z) - z^{-1}Y(z) = 12X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{12}{1 - z^{-1}}$$

Using the z-transform pairs table to convert $H(z)$ to $h(n)$

Table 1 z-transform pairs

Sequence $x(n)$	z-transform $X(z)$	Region of Convergence (ROC)
$\delta(n)$	1	All z
$u(n)$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
$a^n u(n)$	$\frac{1}{1 - az^{-1}}$	$ z > a $
$-a^n u(-n - 1)$	$\frac{1}{1 - az^{-1}}$	$ z < a $
$2 C p ^n \cos(\omega_0 n + \theta)u(n)$	$\frac{C}{1 - pz^{-1}} + \frac{C^*}{1 - p^* z^{-1}}$	$ z > p $

Impulse Response $h(n)$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{12}{1 - z^{-1}}$$

$$h(n) = 12u(n)$$

Standard Structures of Digital Filters

- FIR

- Direct form
- Cascade form
- Linear-phase form
- Frequency-sampling

- IIR

- Direct form I and II
- Cascade form
- Parallel form

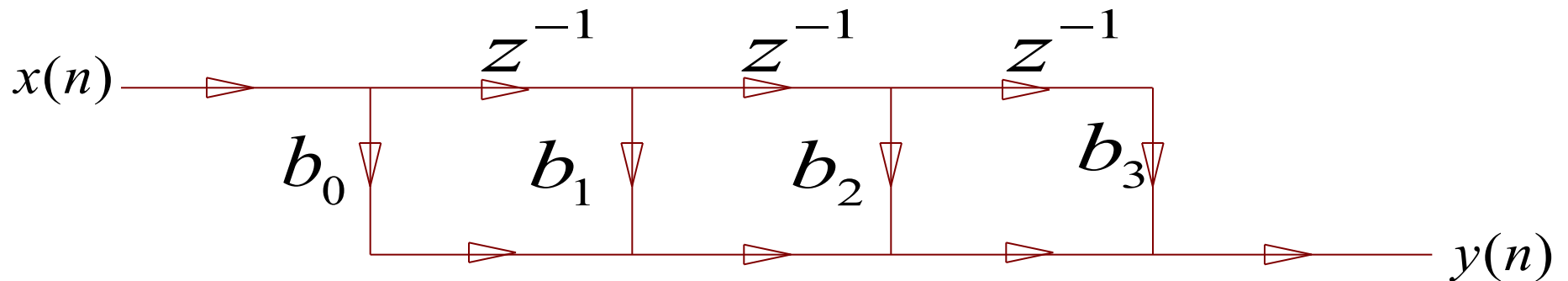
FIR

Direct Form

- FIR equation has the form of

$$H(z) = B(z) = \sum_{k=0}^q b(k)z^{-k}$$

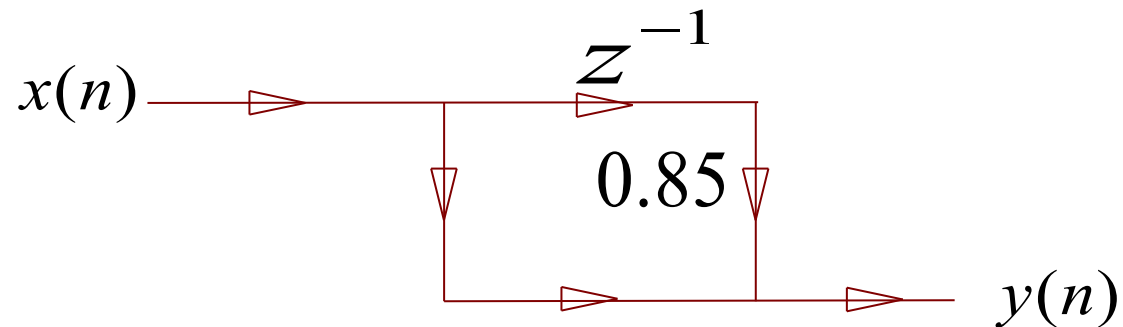
- Example for $q=3$

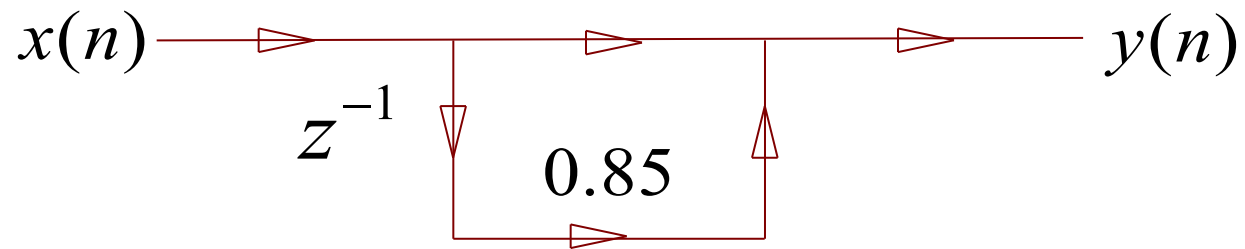
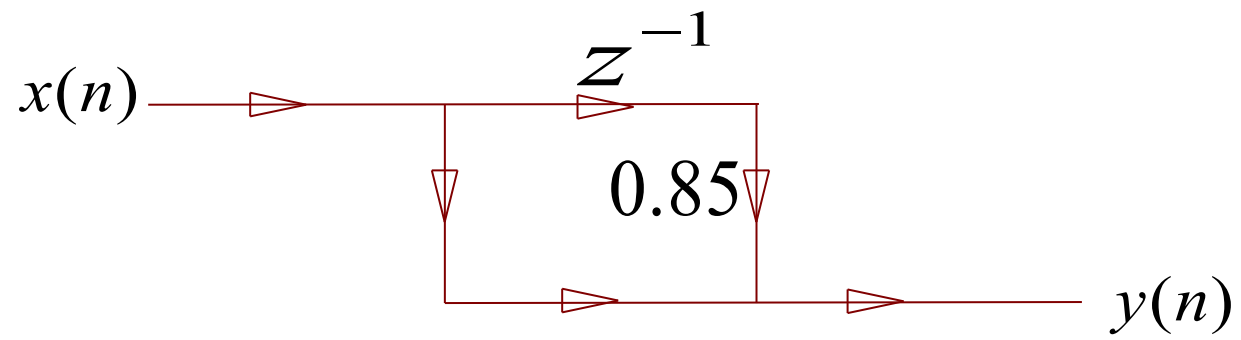


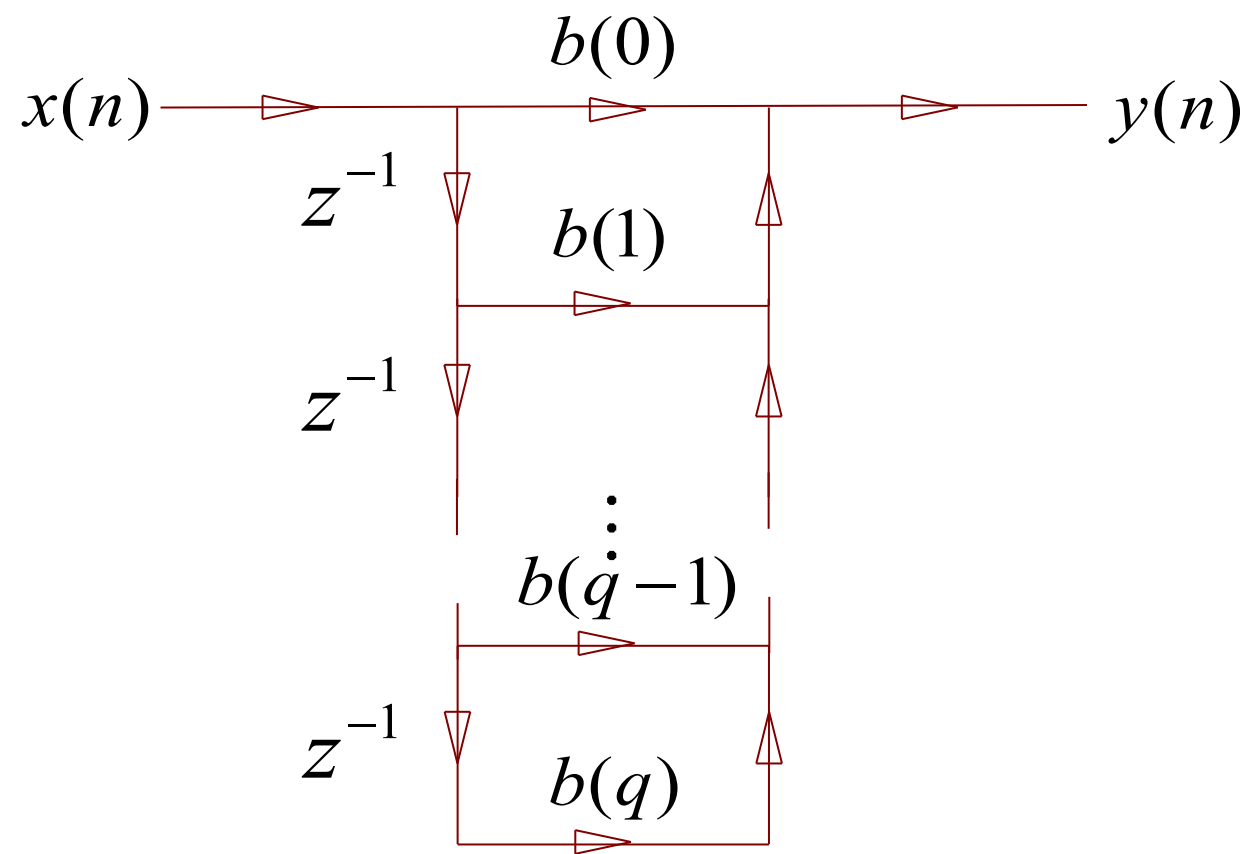
Example 7.7

- Determine the FIR Direct from structure of

$$H(z) = 1 + 0.85z^{-1}$$







IIR: Direct Forms

- IIR equation has the form of

$$H(z) = \frac{B(z)}{A(z)} = \frac{\sum_{k=0}^q b(k)z^{-k}}{1 + \sum_{k=1}^p a(k)z^{-k}}$$

- The corresponding difference equation is

$$y(n) = \sum_{k=0}^q b(k)x(n-k) - \sum_{k=1}^p a(k)y(n-k)$$

- From

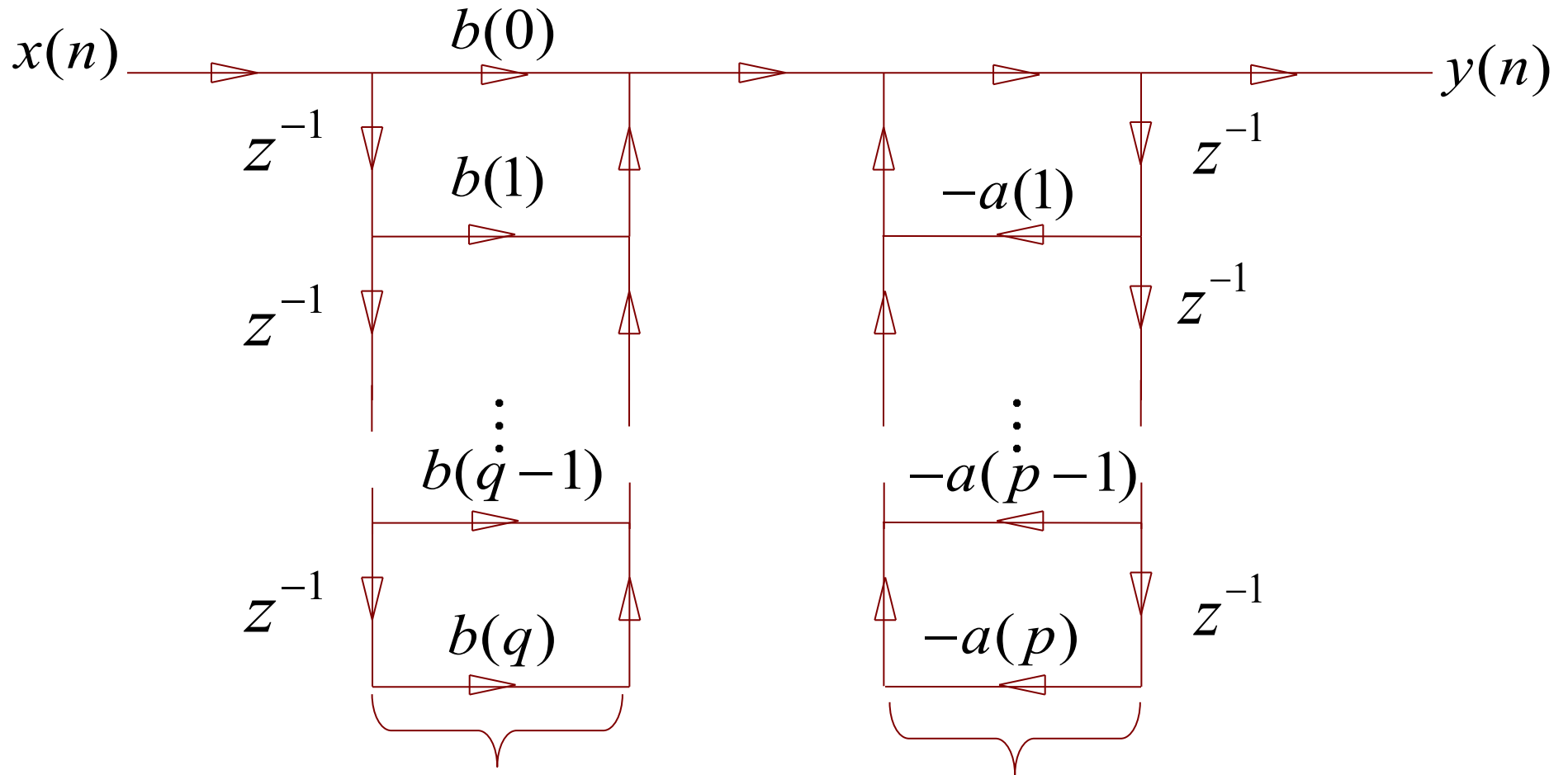
$$y(n) = \sum_{k=0}^q b(k)x(n-k) - \sum_{k=1}^p a(k)y(n-k)$$

- The first term on the RHS is

$$w(n) = \sum_{k=0}^q b(k)x(n-k)$$

- which is the FIR part.

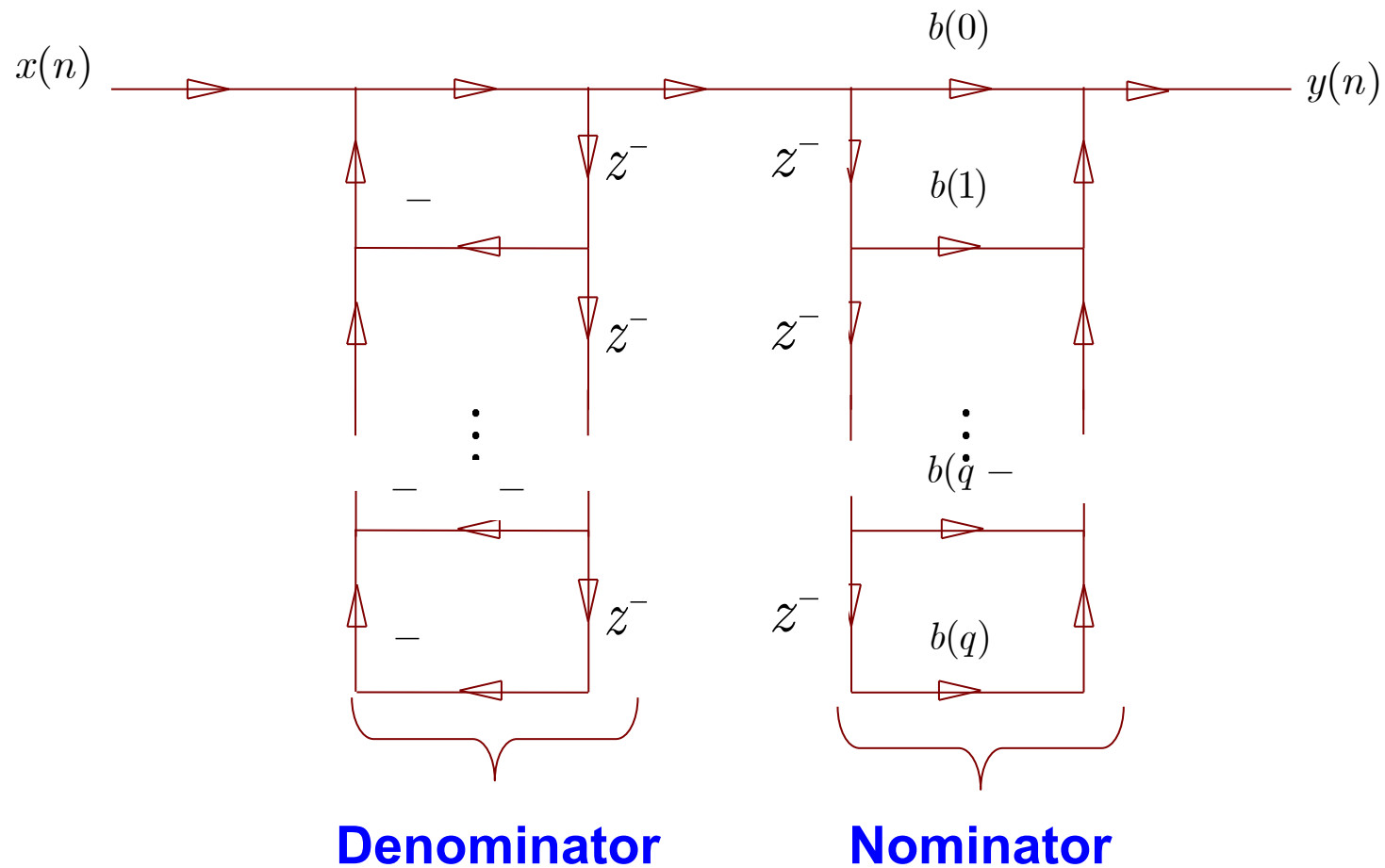
IIR: Direct Form I



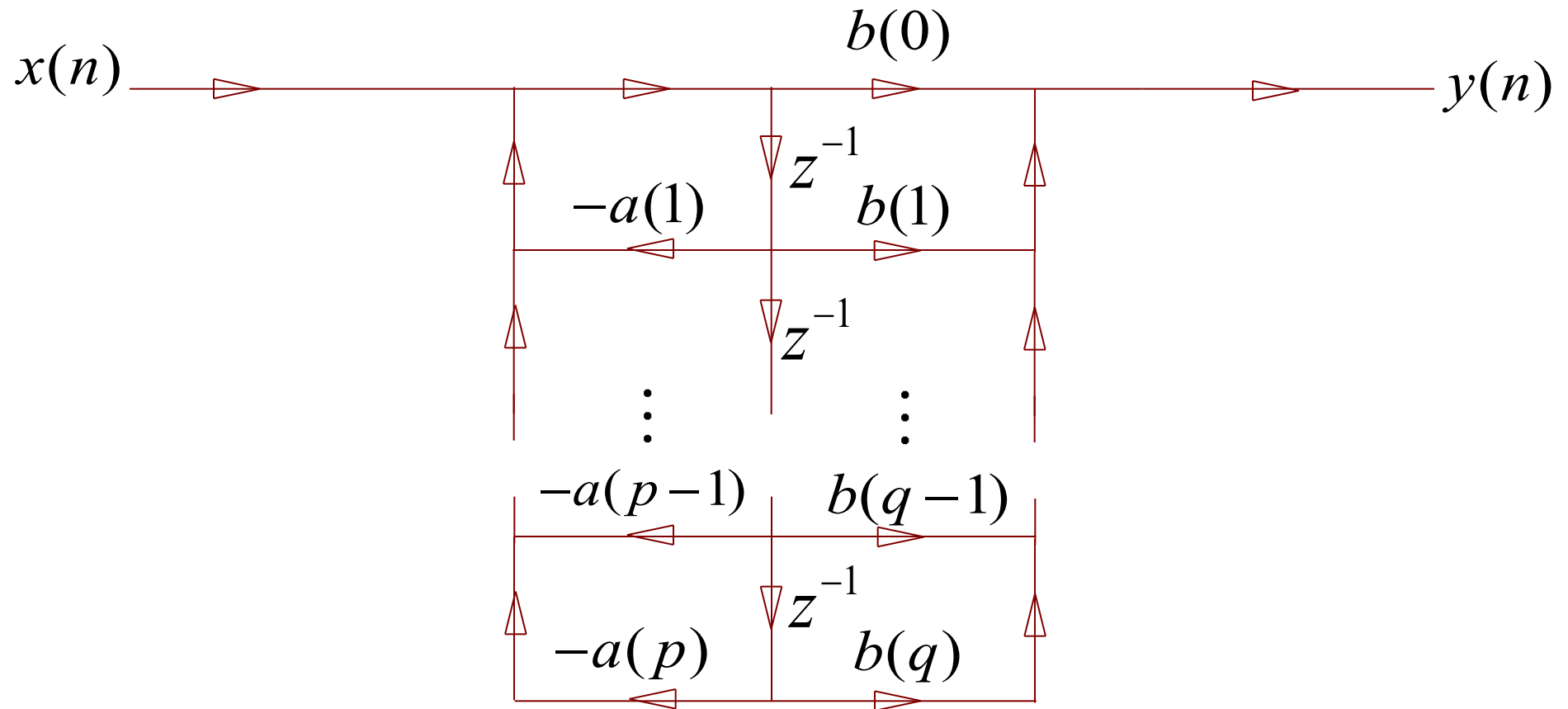
Nominator

Denominator

Interchanging the denominator and nominator



IIR: Direct Form II



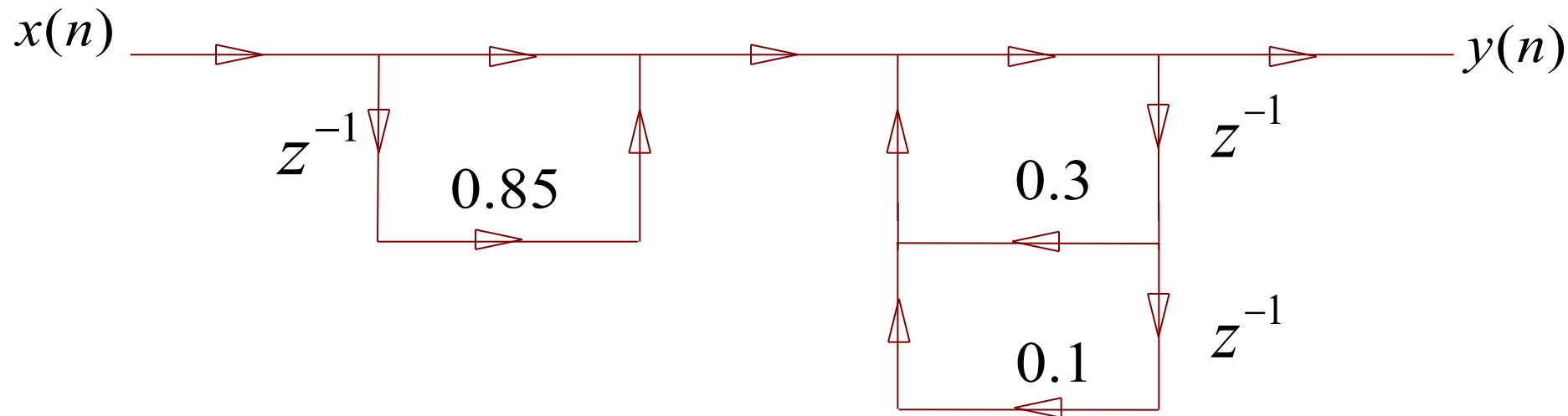
Example 7.8 IIR 1

- Consider the transfer function below and construct the direct form I and II IIR filters.

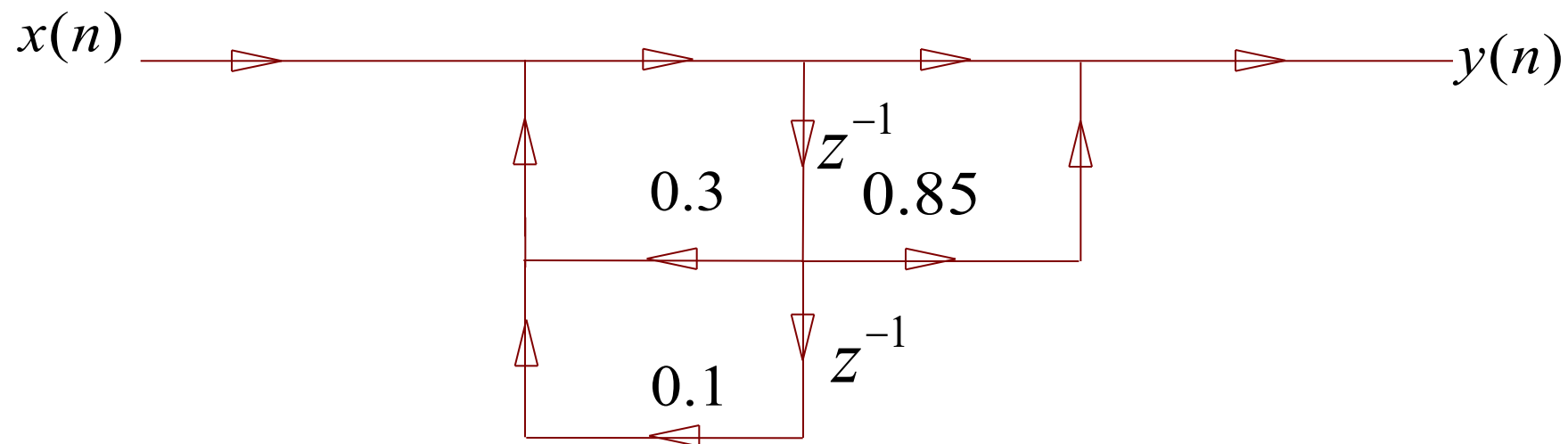
$$H(z) = \frac{1 + 0.85z^{-1}}{(1 + 0.2z^{-1})(1 - 0.5z^{-1})}$$

$$H(z) = \frac{1 + 0.85z^{-1}}{1 - 0.3z^{-1} - 0.1z^{-2}}$$

Direct form I



Direct form II



Homework 7.2

Consider a filter with difference equation

$$2y(n) = x(n) - 5x(n-4) + y(n-1) - y(n-3)$$

Construct the Direct Form II IIR structure of this filter.

Example 7.8 Digital Filter and Frequency Response

- Consider the difference equation

$$y(n) = 0.5x(n) + 0.5x(n-1)$$

$$Y(z) = 0.5X(z) + 0.5z^{-1}X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = 0.5 + 0.5z^{-1}$$

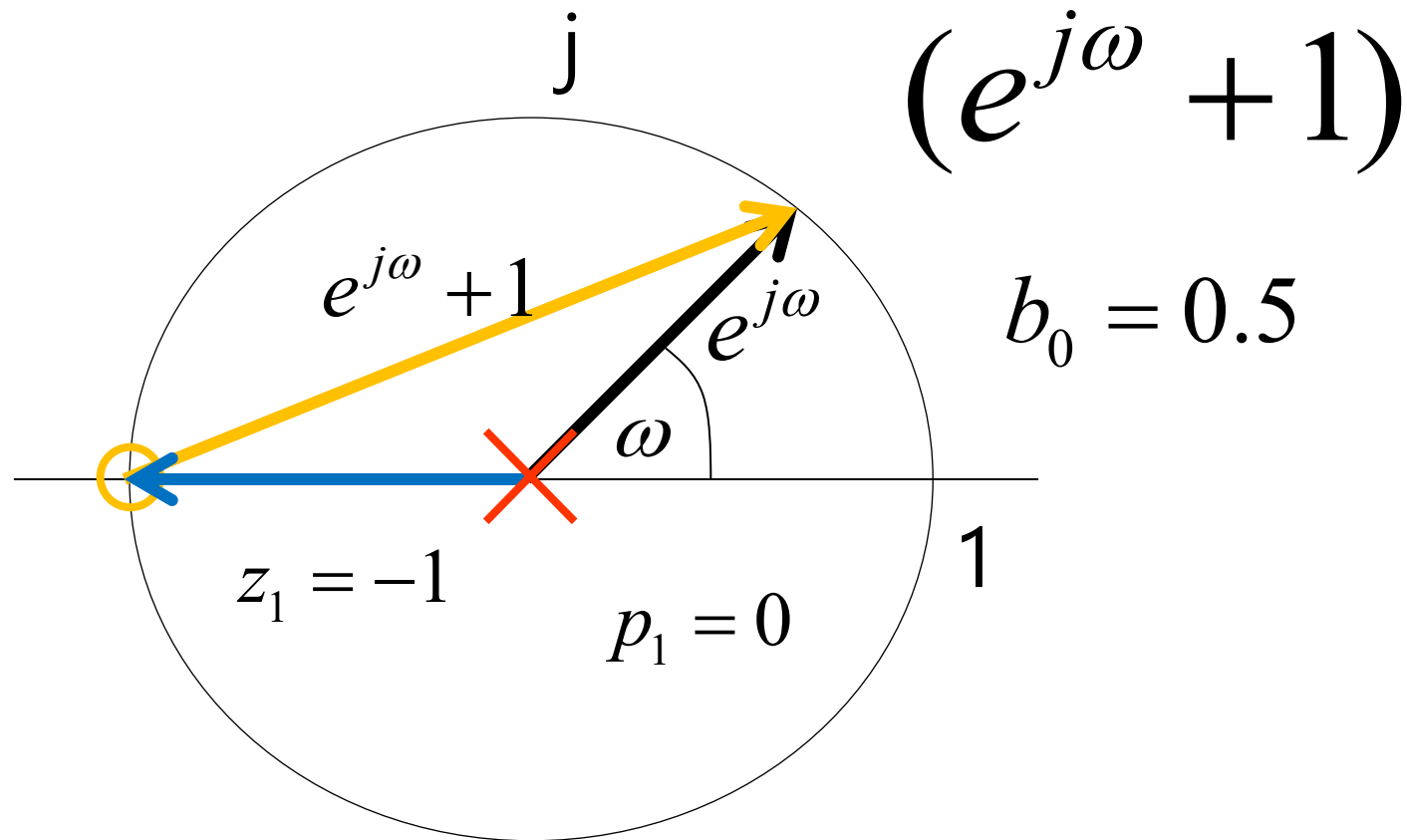
$$H(z) = 0.5 + 0.5z^{-1}$$

- From $z = e^{j\omega}$

$$H(e^{j\omega}) = 0.5 + 0.5e^{-j\omega}$$

$$= 0.5 \frac{e^{j\omega} - (-1)}{e^{j\omega} - 0}$$

$$= b_0 \frac{(e^{j\omega} - z_1)}{(e^{j\omega} - p_1)}$$



$$H(e^{j(0)}) = 0.5 \frac{e^{j(0)} + 1}{e^{j(0)}} = 0.5(1 + 1) = 1 \angle 0$$

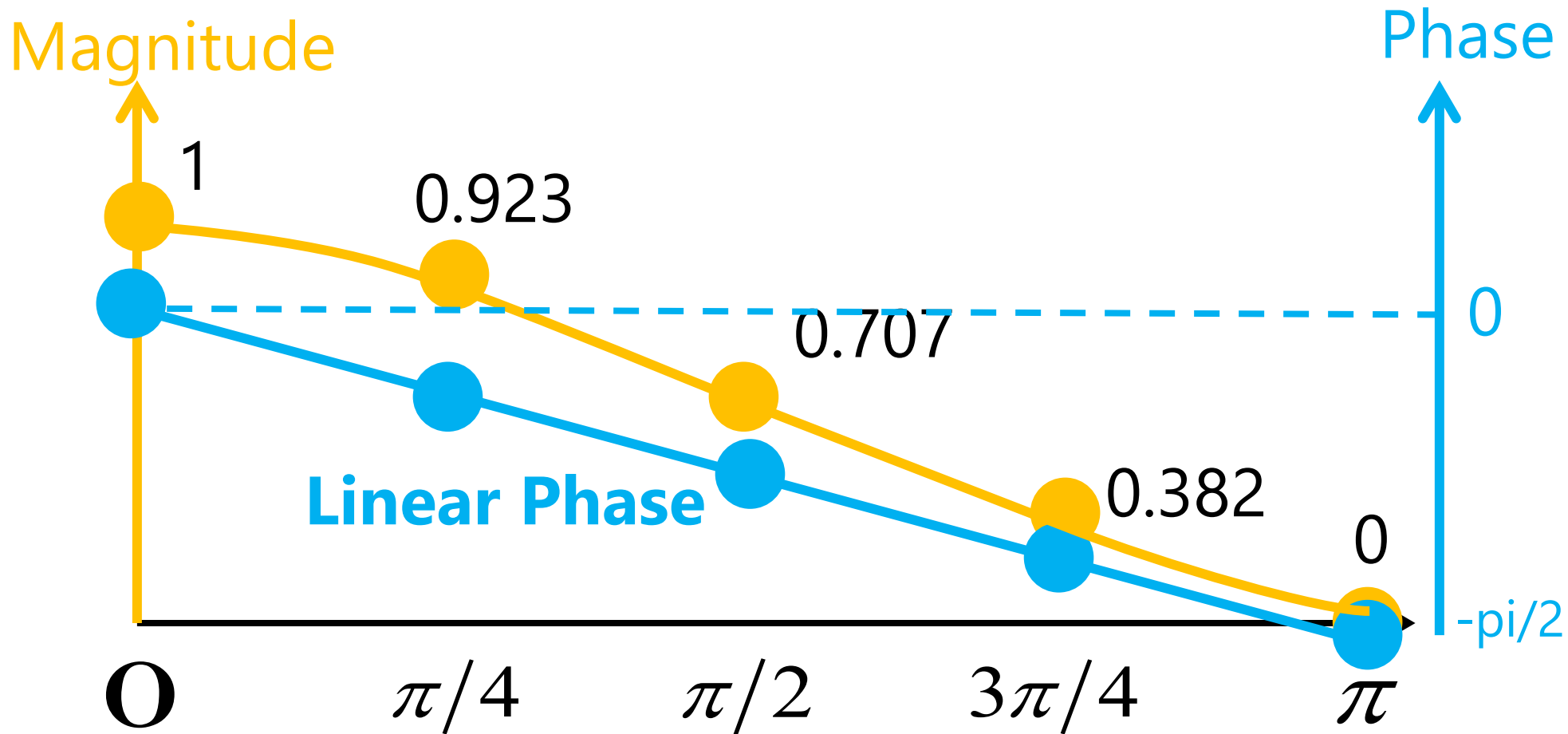
$$H(e^{j(\pi/4)}) = 0.5 \frac{e^{j(\pi/4)} + 1}{e^{j(\pi/4)}} = 0.923 \angle -0.392$$

$$H(e^{j(\pi/2)}) = 0.5 \frac{e^{j(\pi/2)} + 1}{e^{j(\pi/2)}} = 0.707 \angle -0.785$$

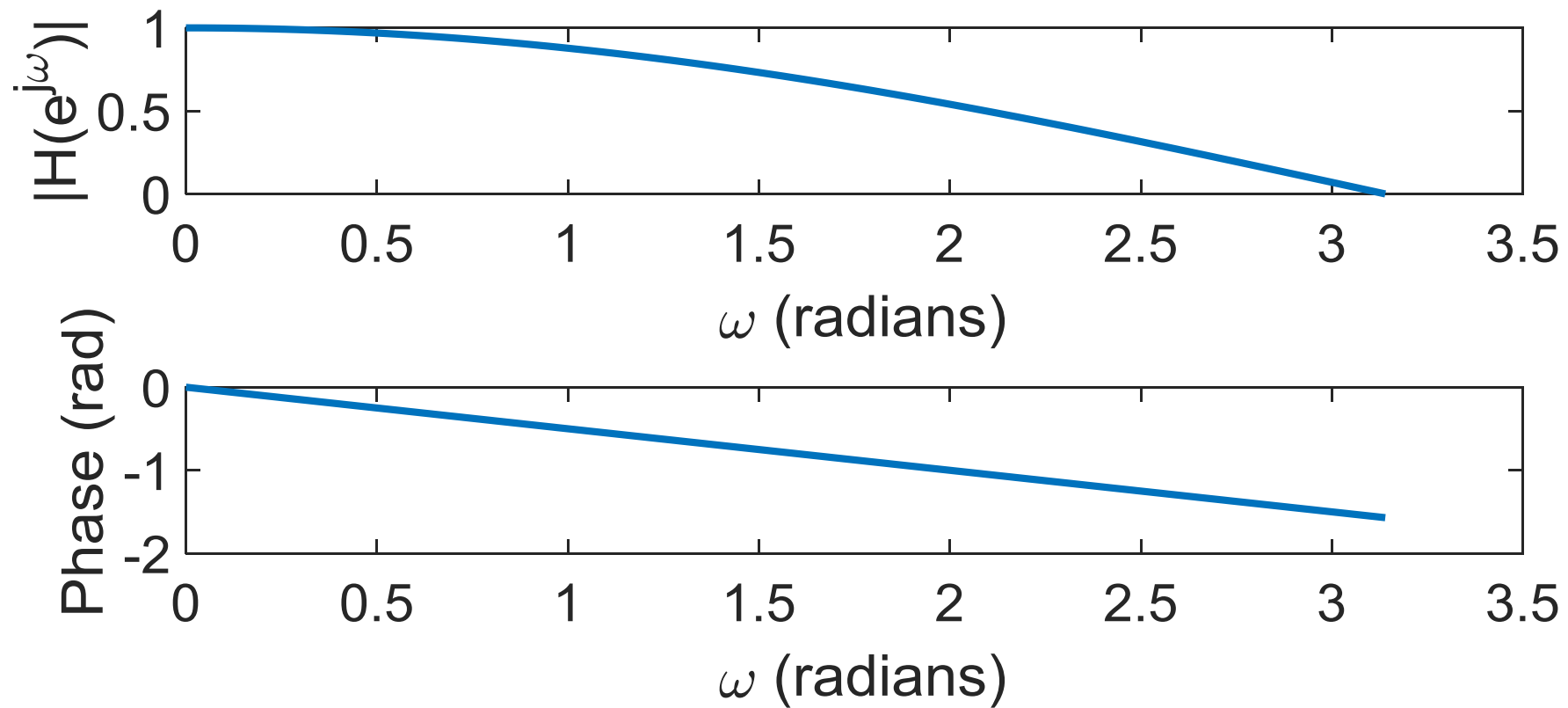
$$H(e^{j(3\pi/4)}) = 0.5 \frac{e^{j(3\pi/4)} + 1}{e^{j(3\pi/4)}} = 0.382 \angle -1.178$$

$$H(e^{j(\pi)}) = 0.5 \frac{e^{j(\pi)} + 1}{e^{j(\pi)}} = 0 \angle -1.57$$

Magnitude and Phase Responses



Magnitude and Phase Responses: Matlab Plots



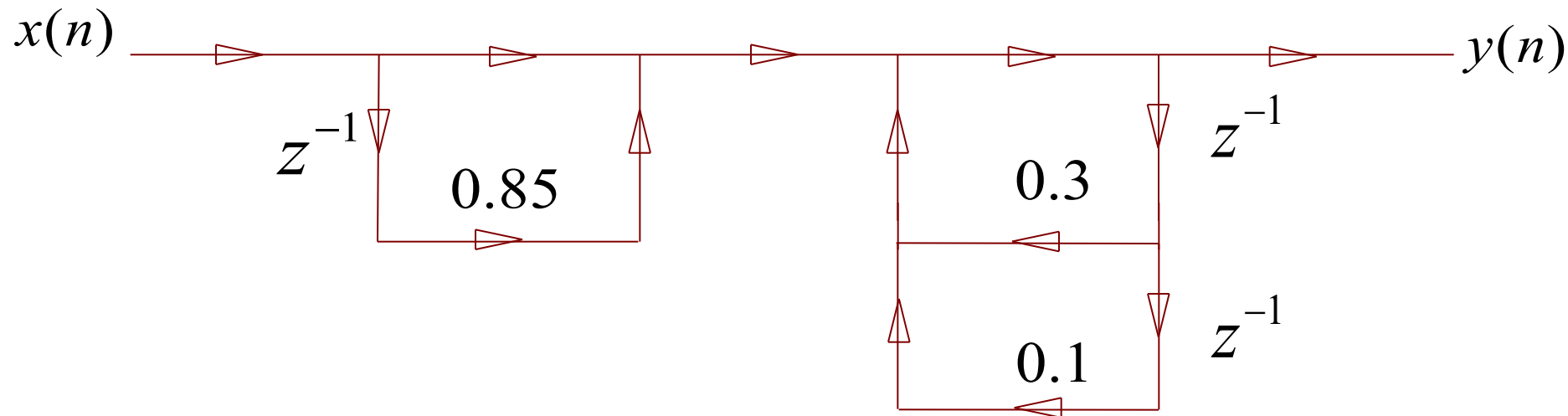
Example 7.9 IIR 1

- Consider the transfer function below and construct the direct form I and II IIR filters.

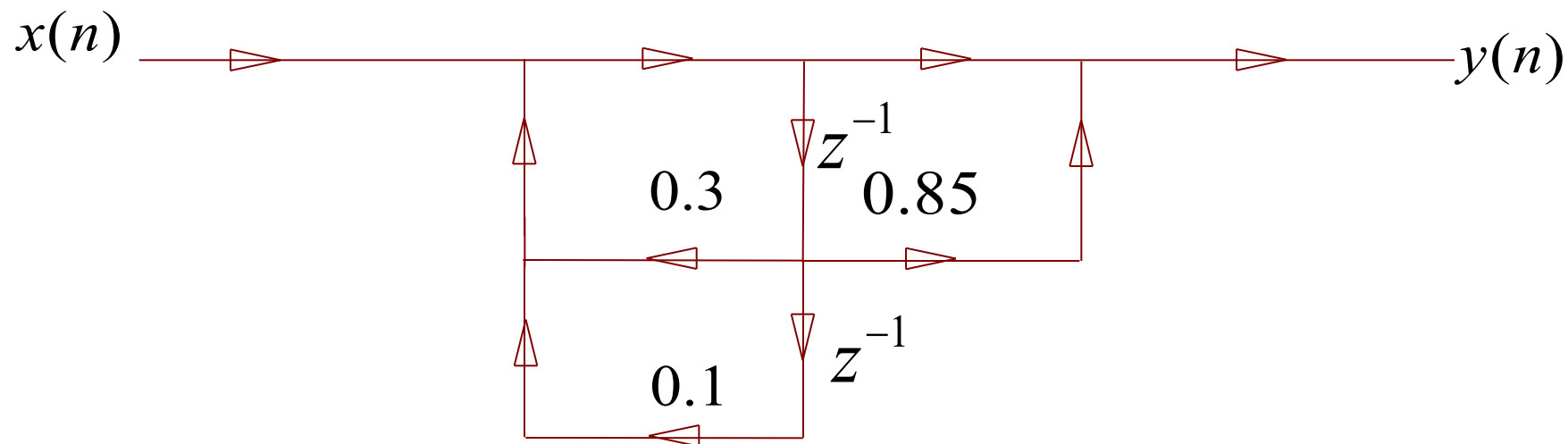
$$H(z) = \frac{1 + 0.85z^{-1}}{(1 + 0.2z^{-1})(1 - 0.5z^{-1})}$$

$$H(z) = \frac{1 + 0.85z^{-1}}{1 - 0.3z^{-1} - 0.1z^{-2}}$$

Direct form I



Direct form II



$$H(z) = \frac{1 + 0.85z^{-1}}{1 - 0.3z^{-1} - 0.1z^{-2}}$$

$$H(e^{j(0)}) = \frac{1 + 0.85e^{-j(0)}}{1 - 0.3e^{-j(0)} - 0.1e^{-j2(0)}} = 3.08 \angle 0$$

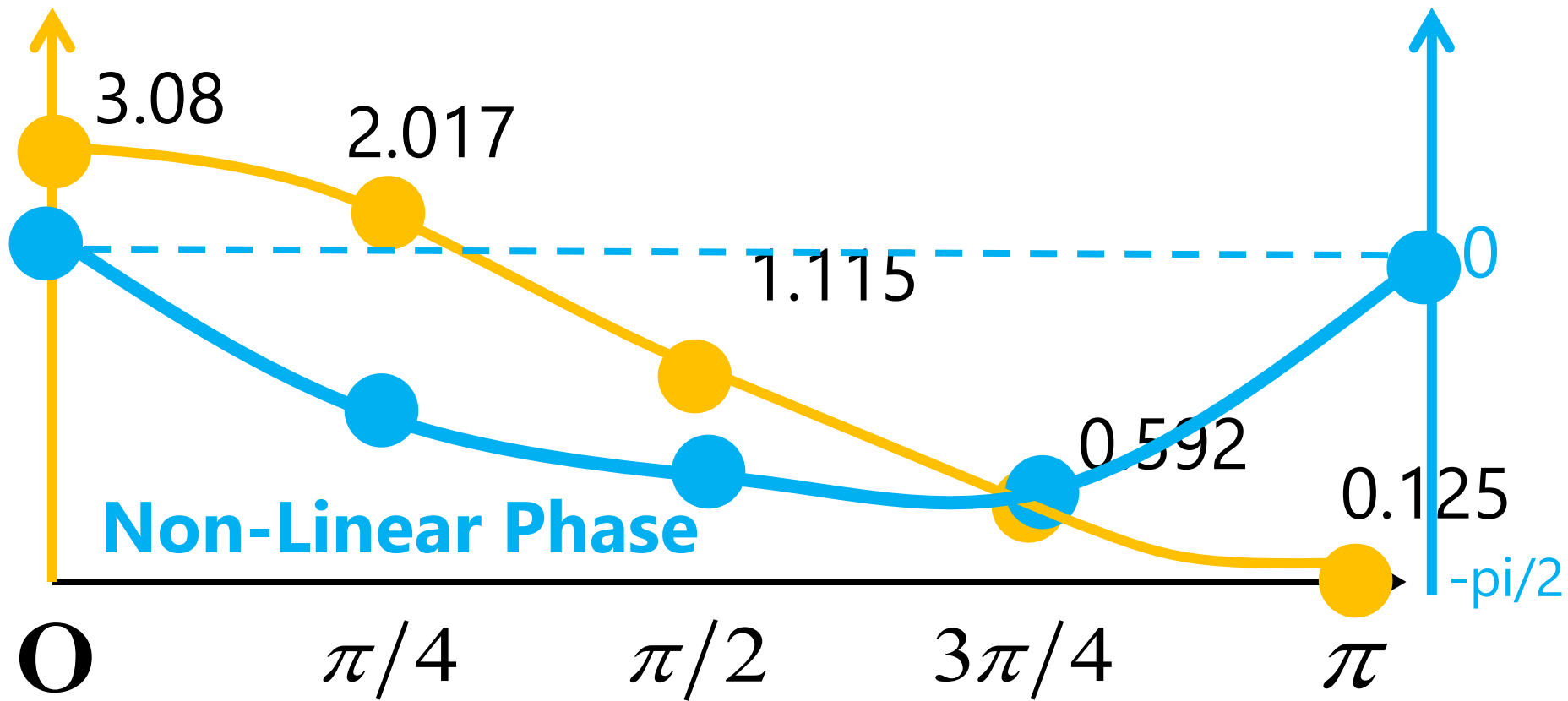
$$H(e^{j(\pi/4)}) = \frac{1 + 0.85e^{-j(\pi/4)}}{1 - 0.3e^{-j(\pi/4)} - 0.1e^{-j2(\pi/4)}} = 2.017 \angle -0.736$$

$$H(e^{j(\pi/2)}) = \frac{1 + 0.85e^{-j(\pi/2)}}{1 - 0.3e^{-j(\pi/2)} - 0.1e^{-j2(\pi/2)}} = 1.115 \angle -0.97$$

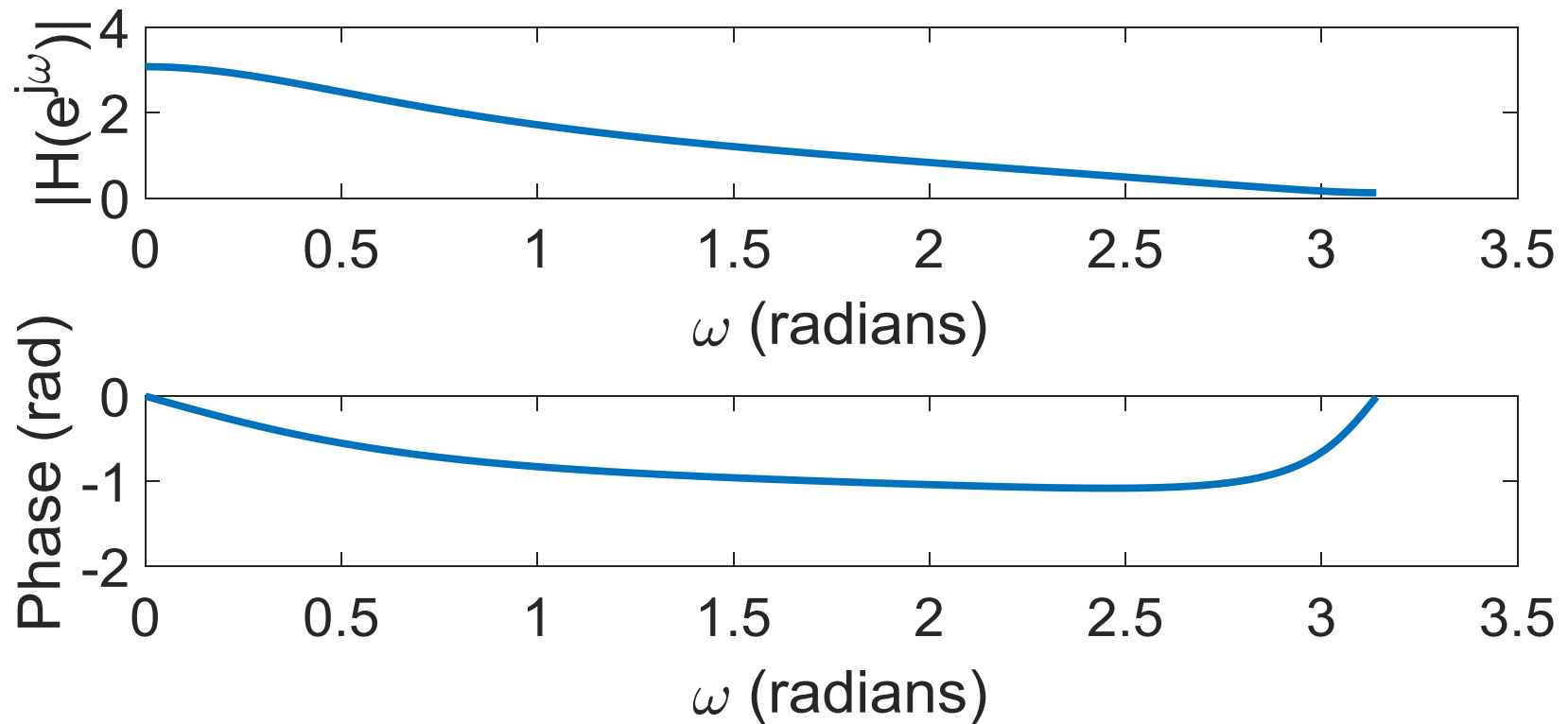
$$H(e^{j(3\pi/4)}) = \frac{1 + 0.85e^{-j(3\pi/4)}}{1 - 0.3e^{-j(3\pi/4)} - 0.1e^{-j2(3\pi/4)}} = 0.592 \angle -1.07$$

$$H(e^{j(\pi)}) = \frac{1 + 0.85e^{-j(\pi)}}{1 - 0.3e^{-j(\pi)} - 0.1e^{-j2(\pi)}} = 0.125 \angle 0$$

Magnitude and Phase Responses



Magnitude and Phase Responses: Matlab plots



Finite Word Length Effects¹

- Converting a floating-point filter to fixed-point can alter the characteristics and performance of the filter significantly.
 - Fixed-point arithmetic can have the following detrimental effects on filter performance.
-
- ¹From http://zone.ni.com/reference/en-XX/help/371325F01/lvdfdtconcepts/dfd_fp_process/#finite_effects

Finite Word Length Effects

- Degraded signal-to-noise ratio (SNR) due to the reduced precision of internal registers, adders, subtracters, and multipliers
- Distorted frequency response from a limited word length representation of filter coefficients
- Overflowed or clipped signal information due to insufficient headroom in the signal paths
- Zero-input limit cycles of infinite impulse response (IIR) filters due to nonlinear quantizers in the feedback loop of IIR filters or to the overflow of the summation operations

FIR: Cascade Form

- The cascade form breaks higher order filter down into several 2nd order filters to alleviate the finite word length effects.

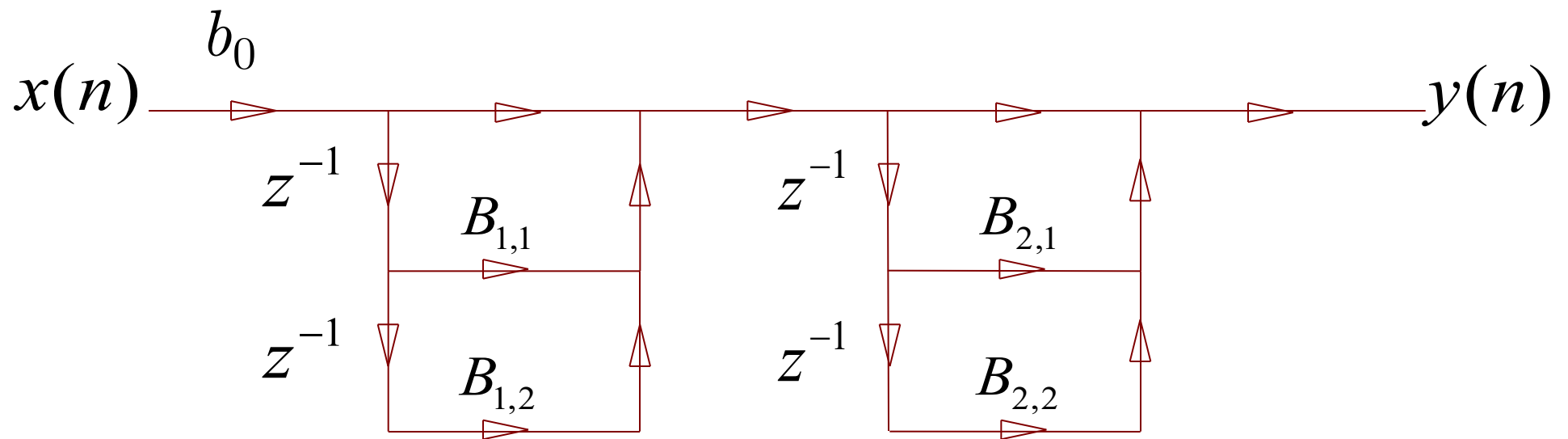
$$H(z) = b_0 + b_1 z^{-1} + \dots + b_{M-1} z^{-M+1}$$

$$= b_0 \left(1 + \frac{b_1}{b_0} z^{-1} + \dots + \frac{b_{M-1}}{b_0} z^{-M+1} \right)$$

$$K = \left\lfloor \frac{M}{2} \right\rfloor \quad = b_0 \prod_{k=1}^K (1 + B_{k,1} z^{-1} + B_{k,2} z^{-2})$$

FIR: Cascade Form

- For $M=5$

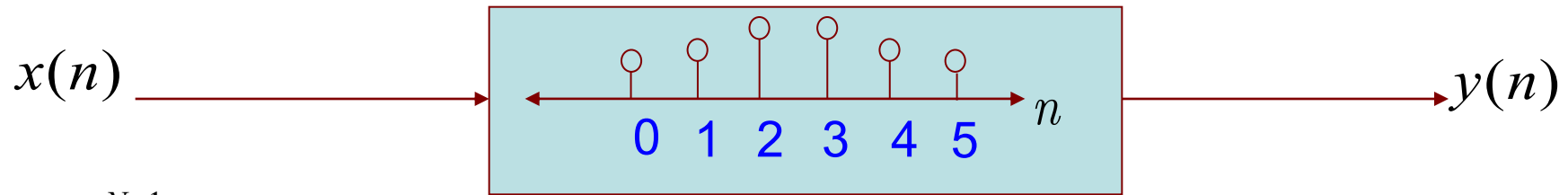


FIR: Linear-phase Form

- For impulse response with length N

Symmertic $h(n) = h(N - 1 - n)$

Anti-Symmertic $h(n) = -h(N - 1 - n)$



$$y(n) = \sum_{k=0}^{N-1} h(k)x(n-k)$$

$$= h(0)x(n) + h(1)x(n-1) + \dots + h(N-2)x(n-N+2) + h(N-1)x(n-N+1)$$

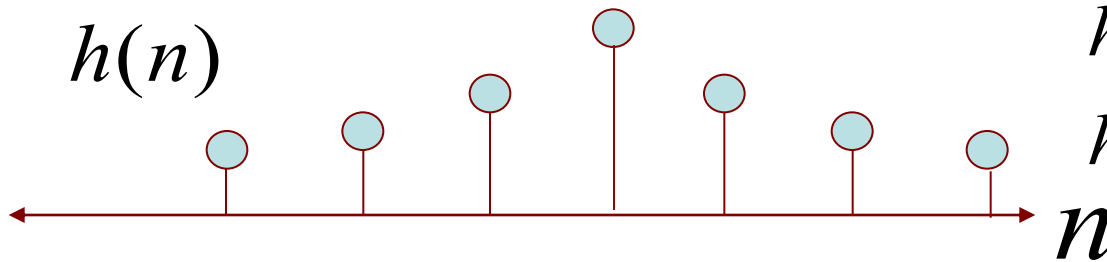
$$= h(0)x(n) + h(1)x(n-1) + \dots + h(1)x(n-N+2) + h(0)x(n-N+1)$$

$$= h(0)\underbrace{[x(n) + x(n-N+1)]} + h(1)\underbrace{[x(n-1) + x(n-N+2)]} \dots$$

This structure reduces the calculation by almost a half.

FIR: Linear-phase Form TYPE I (N = Odd number)

N=7

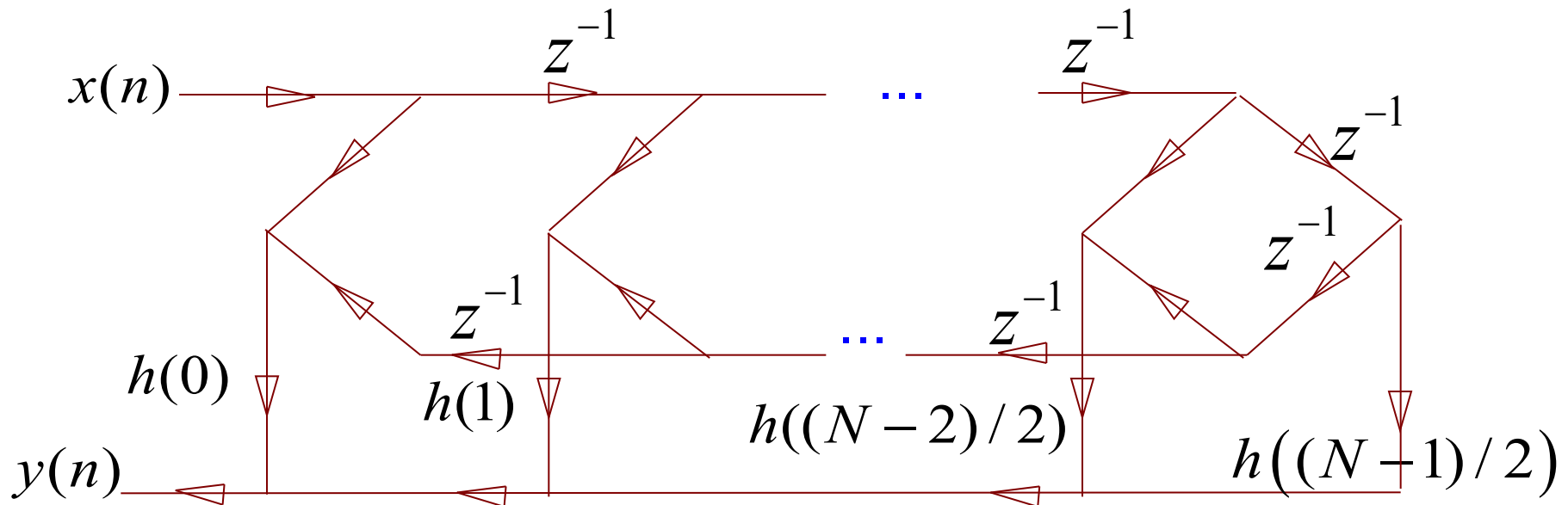


$$h(0) = h((7-1)-0) = h(6)$$

$$h(1) = h((7-1)-1) = h(5)$$

$$h(2) = h((7-1)-2) = h(4)$$

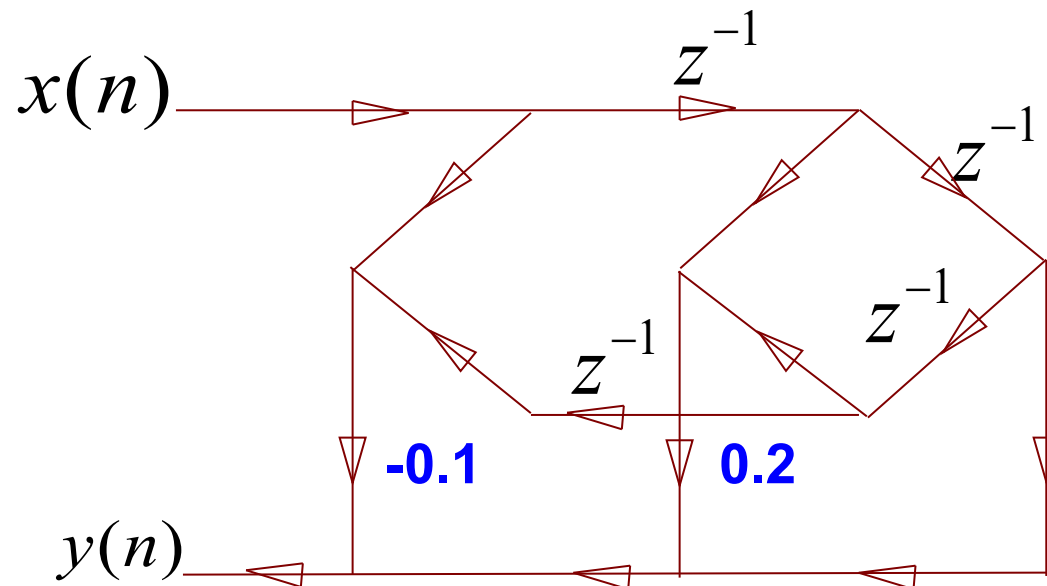
FIR: Linear-phase Form TYPE I (N = Odd number)

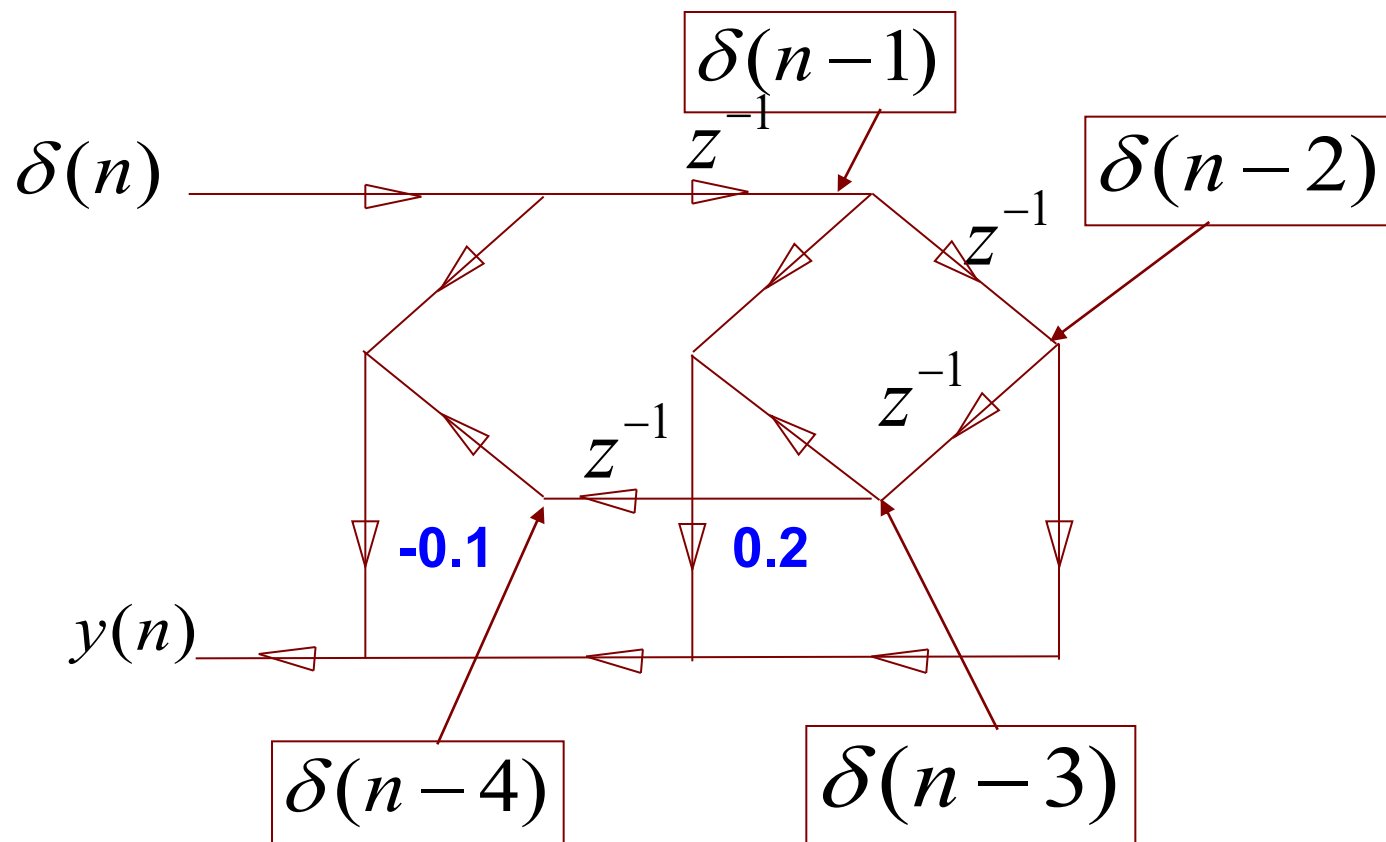


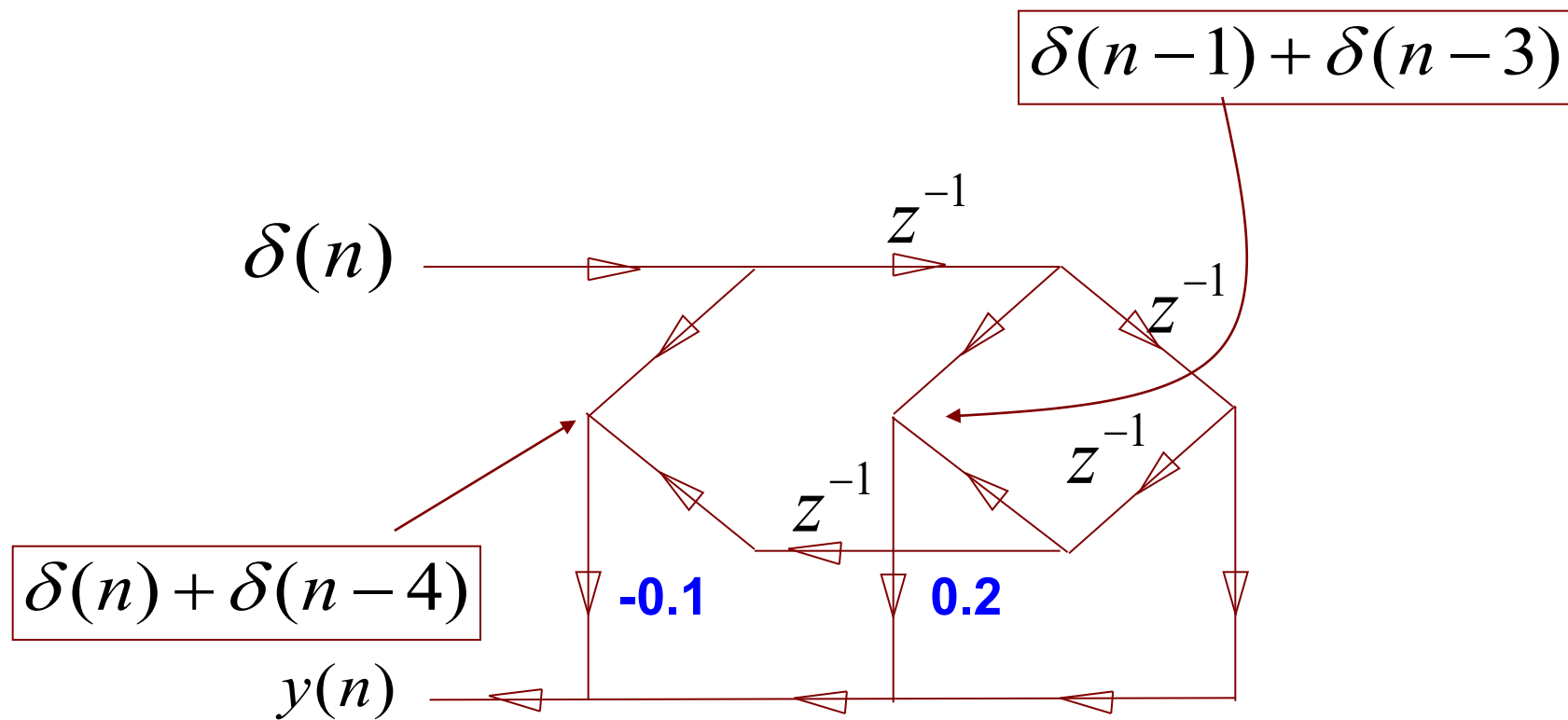
$$y(n) = \sum_{k=0}^{\frac{N-1}{2}} h(k) [x(n-k) + x(n-(N-k-1))] + h\left(\frac{N-1}{2}\right) x\left(n - \frac{N-1}{2}\right)$$

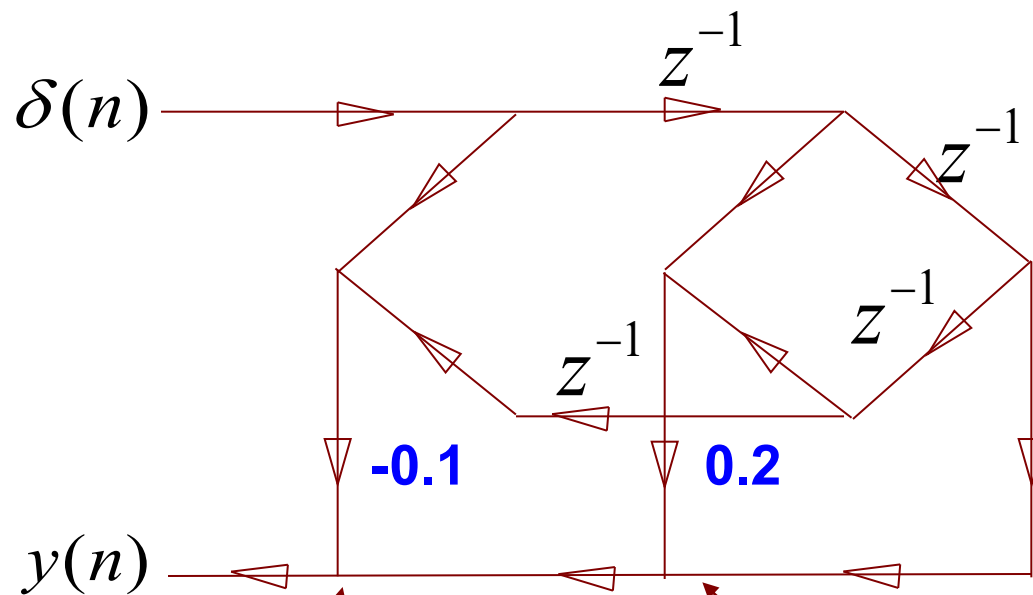
Example 7.10: FIR Linear Phase Type I

- Determine the frequency response of the FIR linear phase type I structure below









$$-0.1[\delta(n) + \delta(n-4)]$$

$$0.2[\delta(n-1) + \delta(n-3)] + \delta(n-2)$$

$$y(n) = -0.1\delta(n) + 0.2\delta(n-1) + \delta(n-2) + 0.2\delta(n-3) - 0.1\delta(n-4)$$

$$H(z) = -0.1 + 0.2z^{-1} + z^{-2} + 0.2z^{-3} - 0.1z^{-4}$$

$$H(e^{j\omega}) = -0.1 + 0.2e^{-j\omega} + e^{-j2\omega} + 0.2e^{-j3\omega} - 0.1e^{-j4\omega}$$

$$H(e^{j0}) = -0.1 + 0.2e^{-j0} + e^{-j2(0)} + 0.2e^{-j3(0)} - 0.1e^{-j4(0)} \\ = 1.2$$

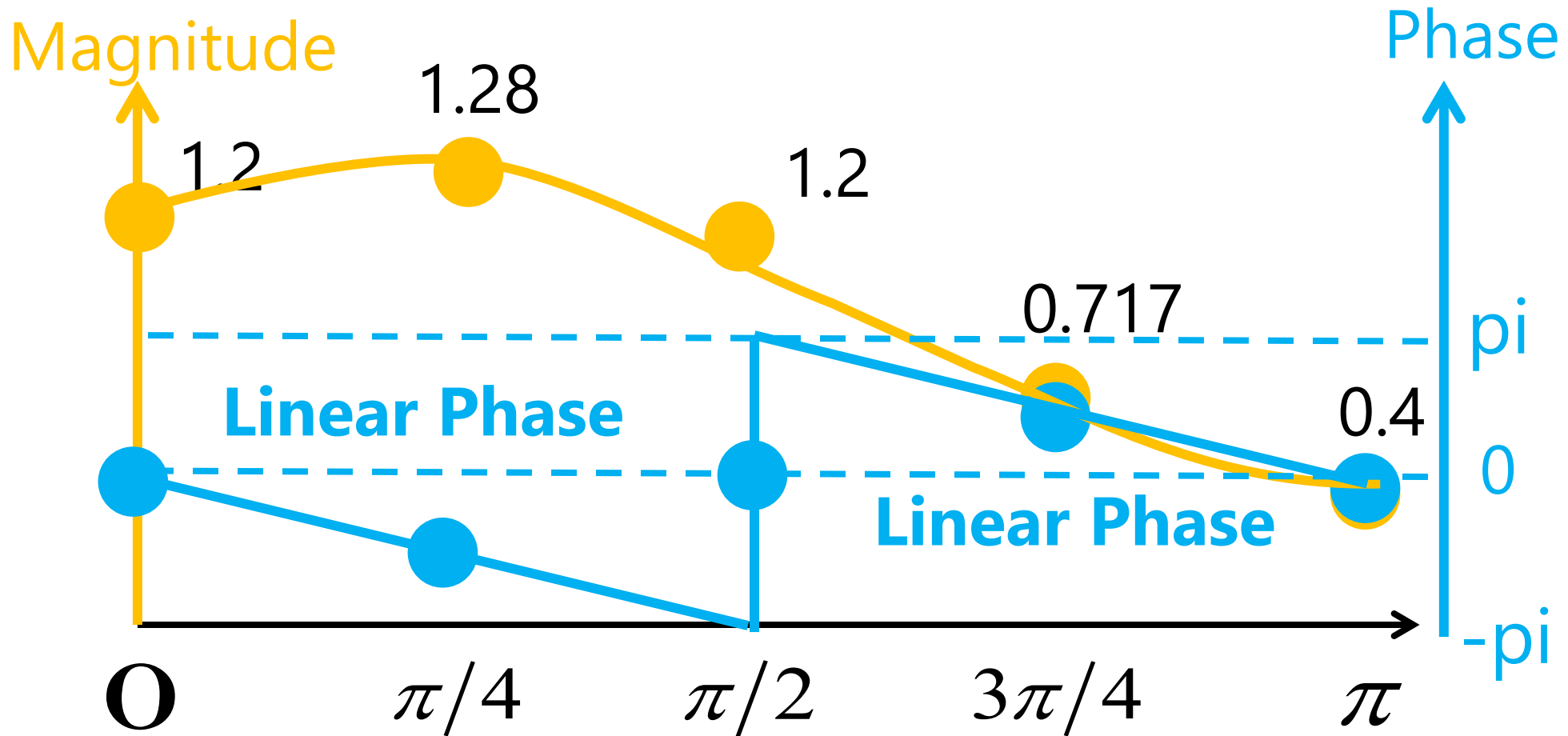
$$H(e^{j\pi/4}) = -0.1 + 0.2e^{-j\pi/4} + e^{-j2(\pi/4)} + 0.2e^{-j3(\pi/4)} - 0.1e^{-j4(\pi/4)} \\ = 1.28 \angle -1.57$$

$$H(e^{j\pi/2}) = -0.1 + 0.2e^{-j\pi/2} + e^{-j2(\pi/2)} + 0.2e^{-j3(\pi/2)} - 0.1e^{-j4(\pi/2)} \\ = 1.2 \angle 0$$

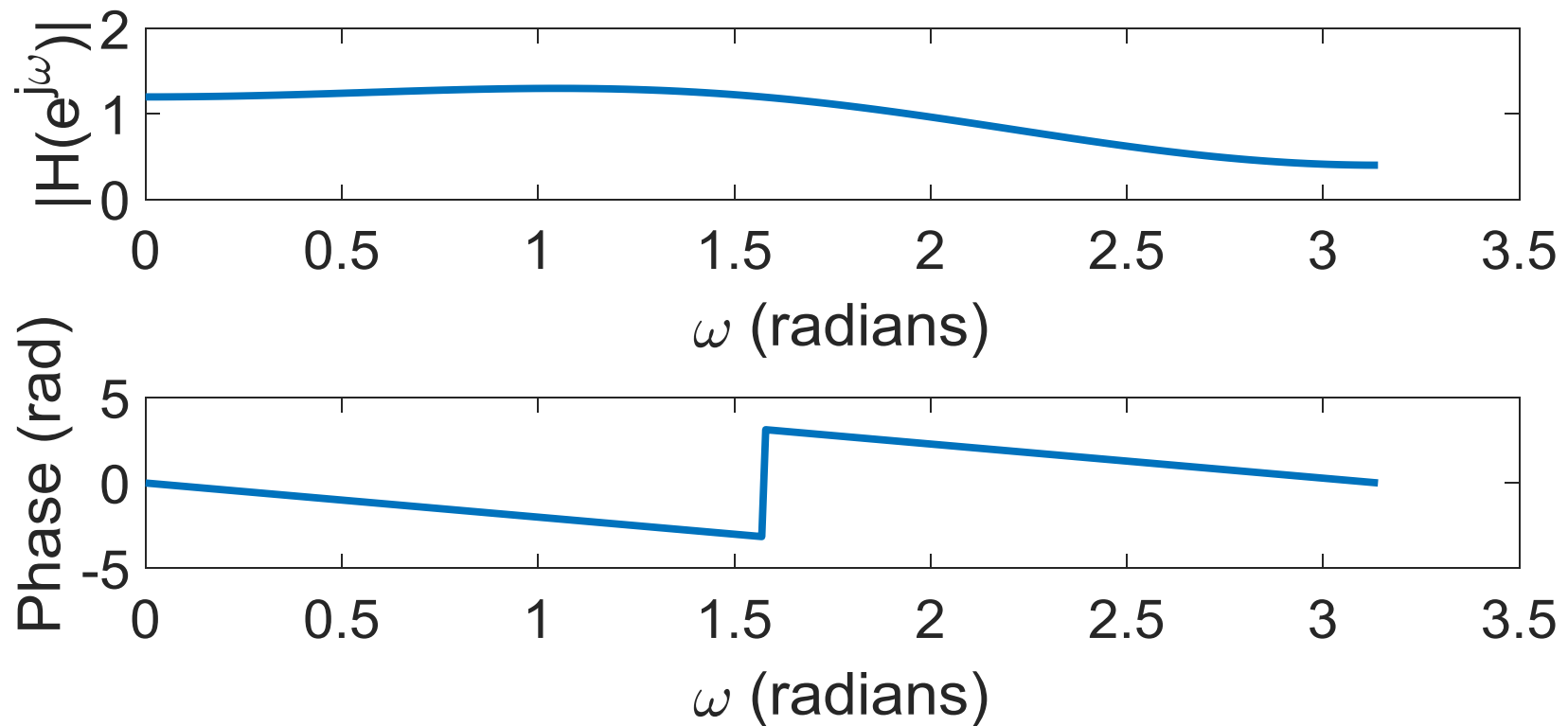
$$H(e^{j3\pi/4}) = -0.1 + 0.2e^{-j3\pi/4} + e^{-j2(3\pi/4)} + 0.2e^{-j3(3\pi/4)} - 0.1e^{-j4(3\pi/4)} \\ = 0.717 \angle 1.57$$

$$H(e^{j\pi}) = -0.1 + 0.2e^{-j\pi} + e^{-j2(\pi)} + 0.2e^{-j3(\pi)} - 0.1e^{-j4(\pi)} \\ = 0.4 \angle 0$$

Magnitude and Phase Response



Magnitude and Phase Responses: Matlab Plots



Example 7.11

- Consider the difference equation

$$y(n) = 0.5x(n) + 1x(n-1) + 0.5x(n-2)$$

- Construct the digital filter structure and plot its frequency response at

$$\omega = 0, \pi/4, \pi/2, 3\pi/4, \pi \quad \text{radians}$$

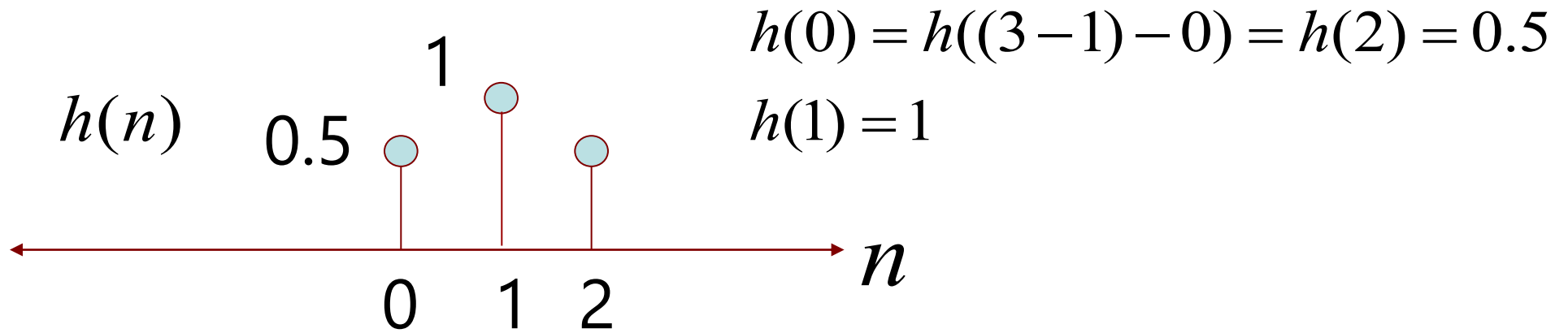
- Solution

$$Y(z) = 0.5X(z) + z^{-1}X(z) + 0.5z^{-2}X(z)$$

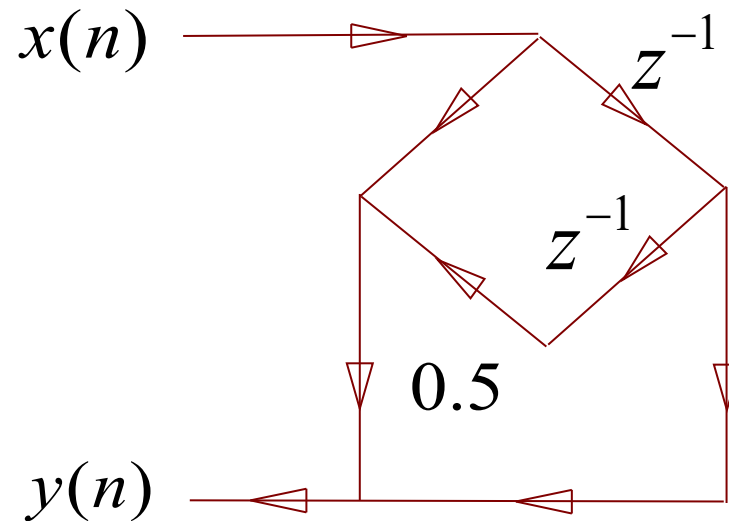
$$H(z) = \frac{Y(z)}{X(z)} = 0.5 + z^{-1} + 0.5z^{-2}$$

$$H(z) = 0.5 + z^{-1} + 0.5z^{-2}$$

- FIR Linear Phase Type I (N=3)



- The digital filter structure of this FIR Linear Phase Type I is



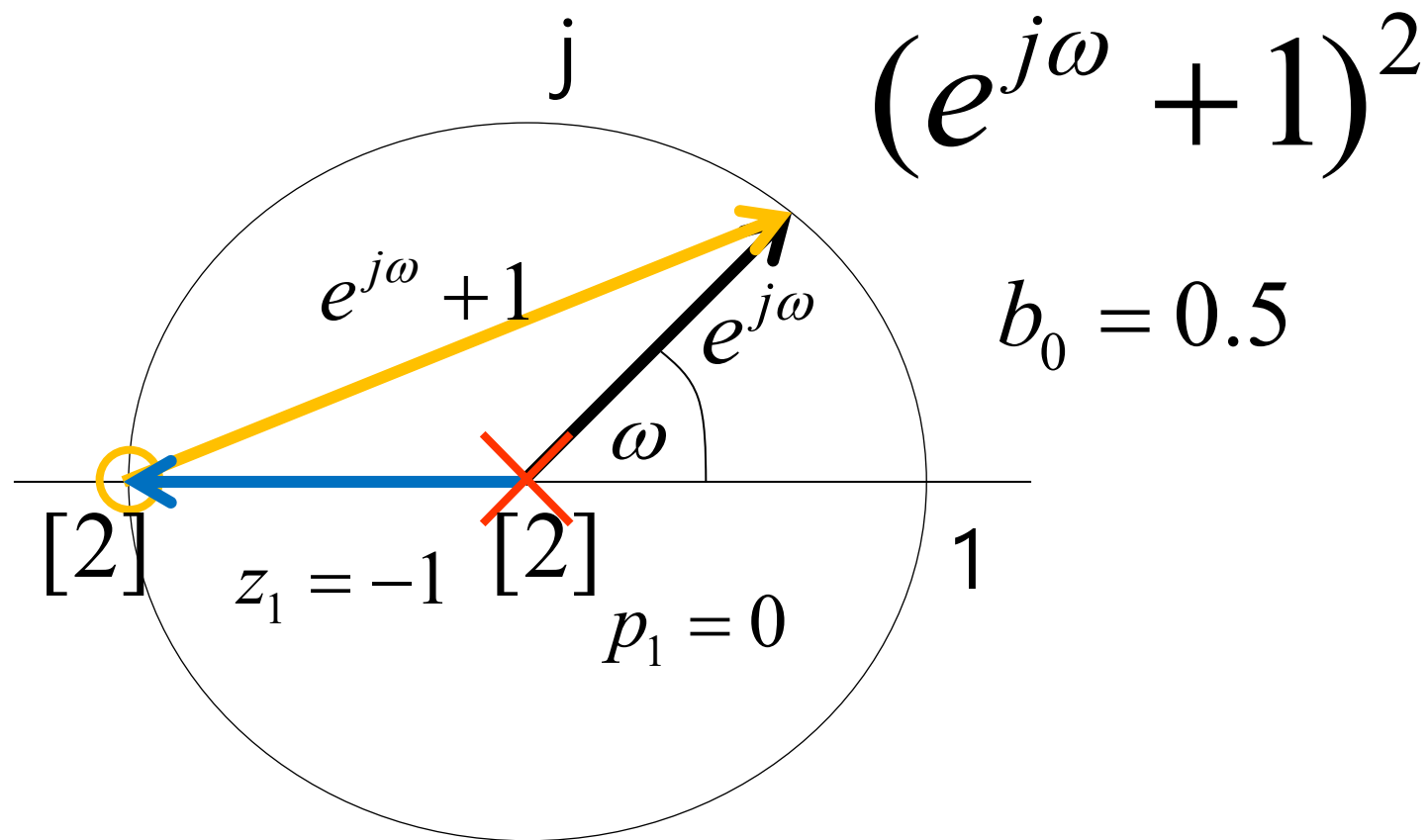
Frequency Response

- From $z = e^{j\omega}$

$$H(e^{j\omega}) = 0.5 + e^{-j\omega} + 0.5e^{-j2\omega}$$

$$= 0.5 \frac{e^{j2\omega} + 2e^{j\omega} + 1}{(e^{j\omega} - 0)^2}$$

$$= b_0 \frac{(e^{j\omega} - z_1)^2}{(e^{j\omega} - p_1)^2}$$



$$H(e^{j(0)}) = 0.5 \frac{(e^{j(0)} + 1)^2}{e^{j2(0)}} = 2 \angle 0$$

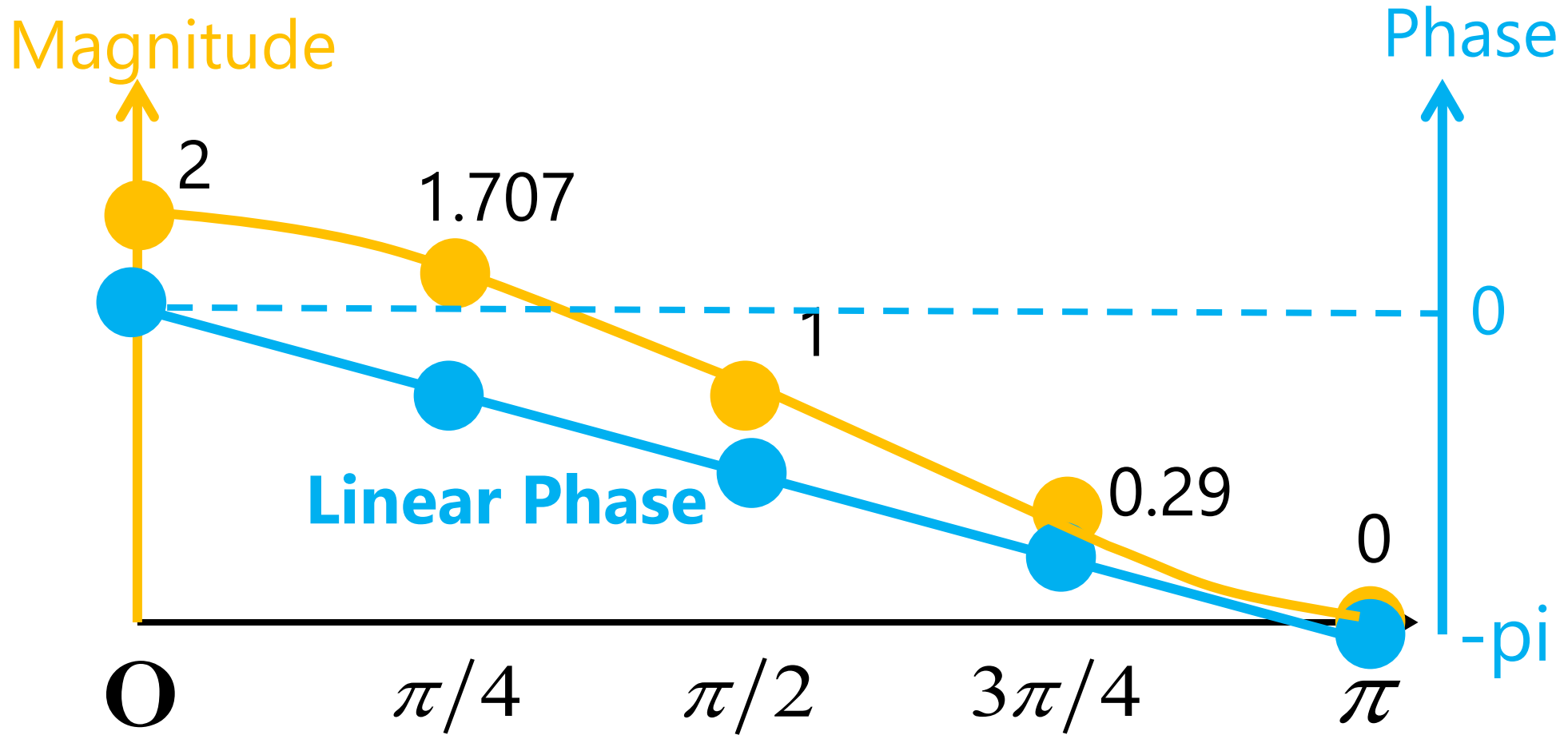
$$H(e^{j(\pi/4)}) = 0.5 \frac{(e^{j(\pi/4)} + 1)^2}{e^{j2(\pi/4)}} = 1.707 \angle -0.785$$

$$H(e^{j(\pi/2)}) = 0.5 \frac{(e^{j(\pi/2)} + 1)^2}{e^{j2(\pi/2)}} = 1 \angle -1.57$$

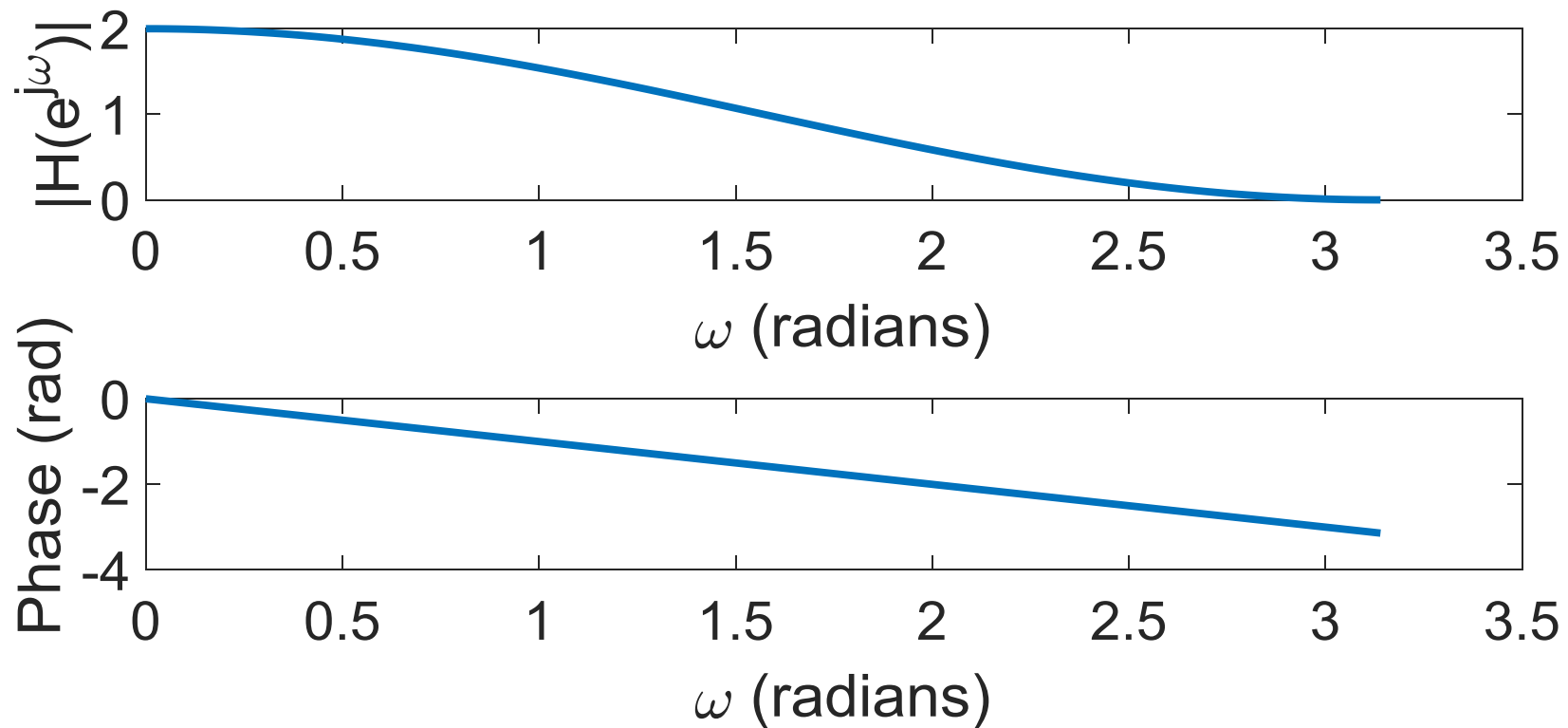
$$H(e^{j(3\pi/4)}) = 0.5 \frac{(e^{j(3\pi/4)} + 1)^2}{e^{j2(3\pi/4)}} = 0.29 \angle -2.35$$

$$H(e^{j(\pi)}) = 0.5 \frac{(e^{j(\pi)} + 1)^2}{e^{j2(\pi)}} = 0 \angle -3.14$$

Magnitude and Phase Responses



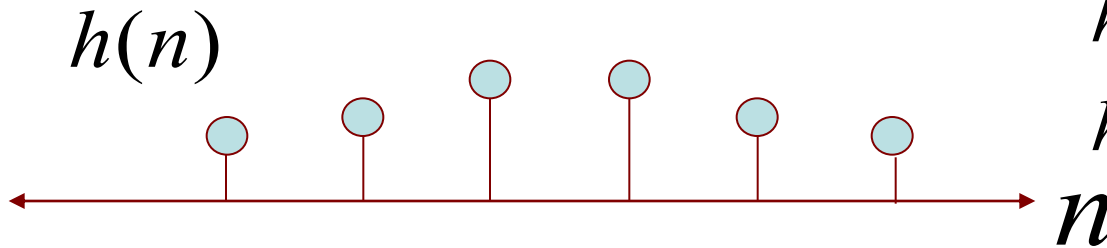
Magnitude and Phase Responses: Matlab Plots



FIR: Linear-phase Form

TYPE II ($N = \text{Even number}$)

- N is an even number and $h(n)$ is symmetric

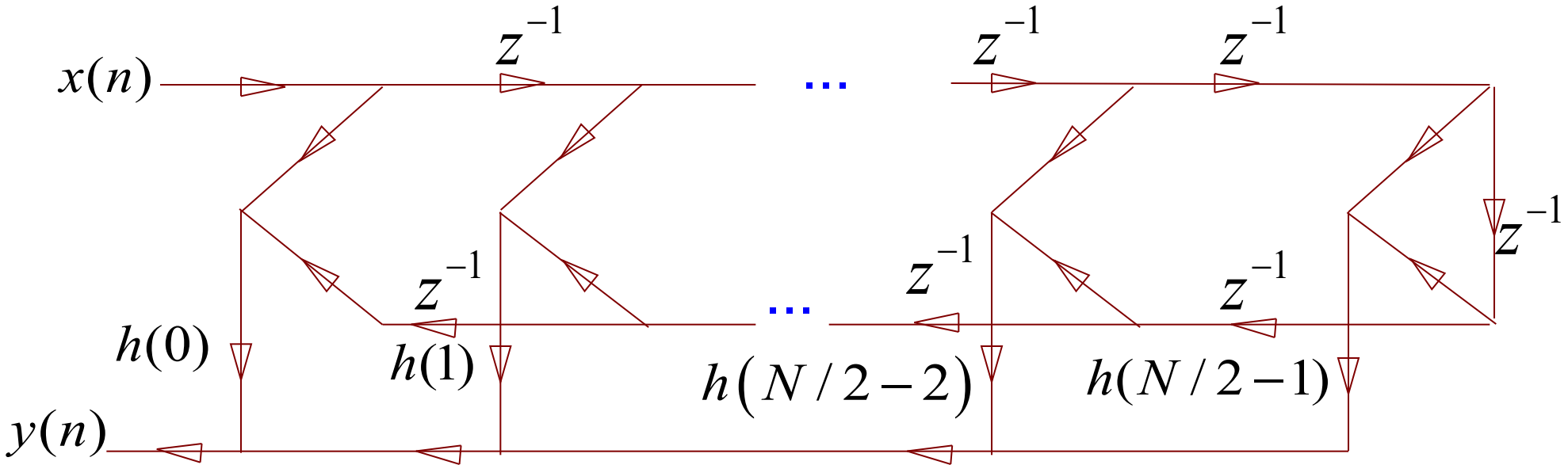


$$h(0) = h((6-1)-0) = h(5)$$

$$h(1) = h((6-1)-1) = h(4)$$

$$h(2) = h((6-1)-2) = h(3)$$

FIR: Linear-phase Form TYPE II (N = Even number)



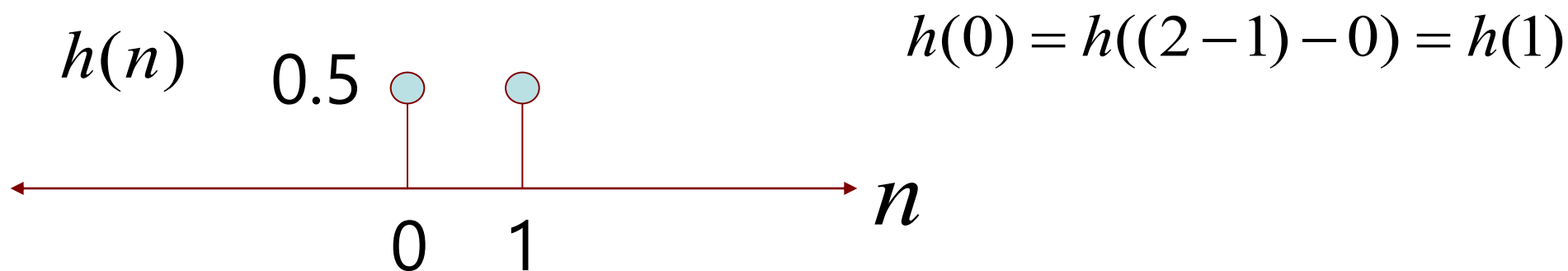
$$y(n) = \sum_{k=0}^{N/2-1} h(k)[x(n-k) + x(n-(N-k-1))]$$

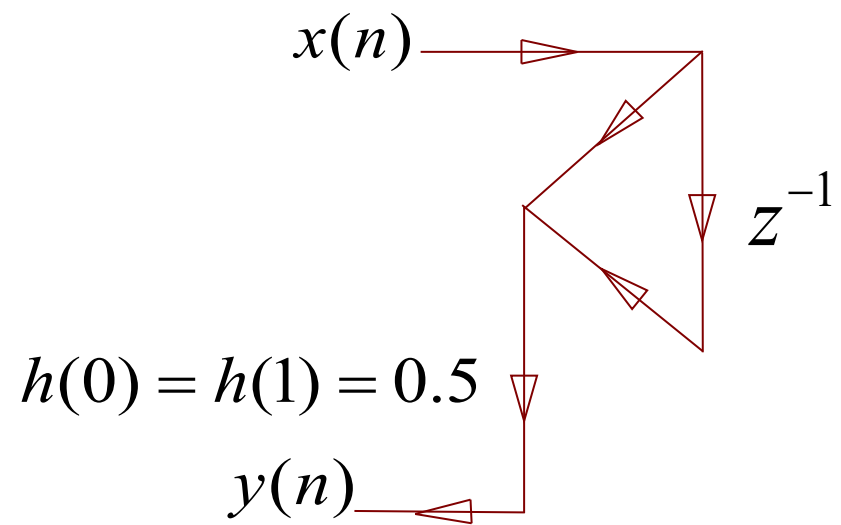
Example 7.11

$$H(z) = 0.5 + 0.5z^{-1}$$

$$h(n) = 0.5 + 0.5\delta(n-1)$$

$$h(n) = h(0) + h(1)\delta(n-1)$$





Homework 7.3

- Consider the difference equation

$$y(n) = x(n) + 2x(n-1) + x(n-2)$$

- Construct the digital filter structure in a **linear phase form** and plot its frequency responses at
at

$$\omega = 0, \pi/4, \pi/2, 3\pi/4, \pi \quad \text{radians}$$

FIR: Frequency Sampling (1)

- Given the impulse response

$$h(n) = \frac{1}{N} \sum_{k=0}^{N-1} H(k) e^{j2\pi nk/N}$$

$$\begin{aligned} H(z) &= \sum_{n=0}^{N-1} h(n) z^{-n} = \sum_{n=0}^{N-1} \left[\frac{1}{N} \sum_{k=0}^{N-1} H(k) e^{j2\pi nk/N} \right] z^{-n} \\ &= \frac{1}{N} \sum_{k=0}^{N-1} H(k) \sum_{n=0}^{N-1} e^{j2\pi nk/N} z^{-n} \end{aligned}$$

FIR: Frequency Sampling (2)

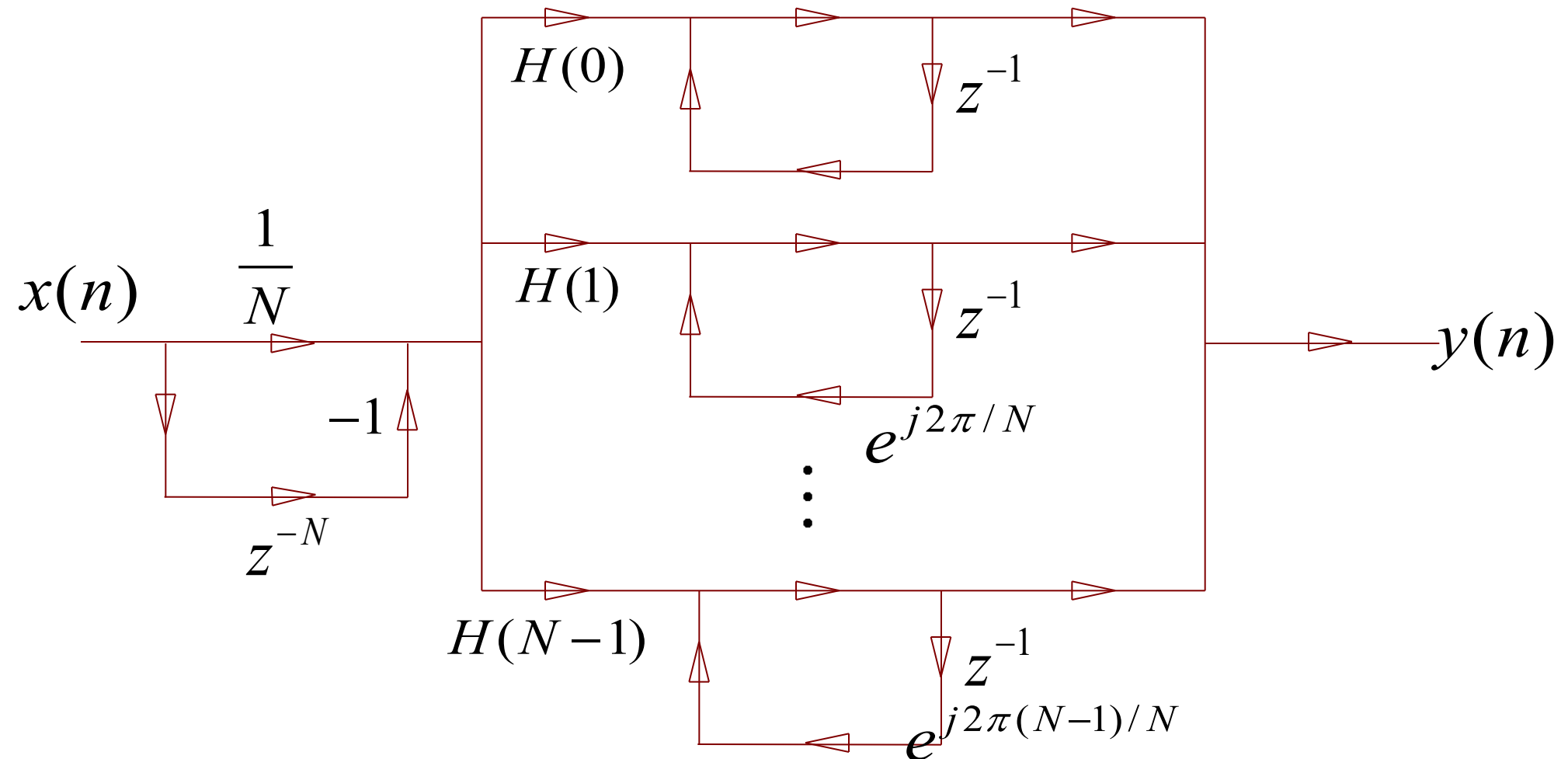
- From
$$H(z) = \frac{1}{N} (1 - z^{-N}) \sum_{k=0}^{N-1} \frac{H(k)}{1 - e^{j2\pi k/N} z^{-1}}$$

- Compare to the cascade structure

$$G(z) = \frac{1}{N} (1 - z^{-N})$$

$$H_k(z) = \frac{H(k)}{1 - e^{j2\pi k/N} z^{-1}}$$

FIR: Frequency Sampling (3)



IIR: Cascade

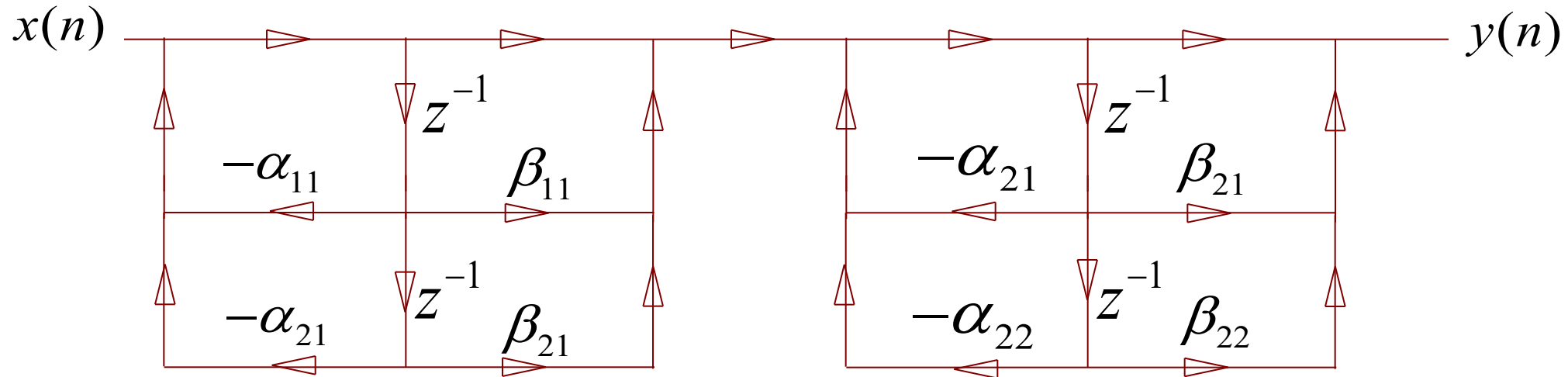
- For first order IIR, we construct the cascade form of

$$H(z) = \frac{\sum_{k=0}^q b(k)z^{-k}}{1 + \sum_{k=1}^p a(k)z^{-k}} = A \prod_{k=1}^{\max(p,q)} \frac{1 - \beta_k z^{-1}}{1 - \alpha_k z^{-1}}$$

- and for second order IIRs

$$H_k(z) = \frac{1 + \beta_{1k}z^{-1} + \beta_{2k}z^{-2}}{1 + \alpha_{1k}z^{-1} + \alpha_{2k}z^{-2}}$$

IIR: Cascade



IIR: Parallel Form

$$H(z) = \frac{\sum_{k=0}^q b(k)z^{-k}}{1 + \sum_{k=1}^p a(k)z^{-k}} = A \frac{\prod_{k=1}^q 1 - \beta_k z^{-1}}{\prod_{k=1}^p 1 - \alpha_k z^{-1}}$$

$$H(z) = \sum_{k=1}^{N_s} \frac{\gamma_{0k} + \gamma_{1k}z^{-1}}{1 + \alpha_{1k}z^{-1} + \alpha_{2k}z^{-2}}$$

γ, α_k = Coefficients

IIR: Parallel Form

- $N=4$ ($N_s=2$)

